State Emergency Management Plan
Bushfire Sub-Plan



VICTORIA
State Government

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Acknowledgment of Country

EMV acknowledges Aboriginal and Torres Strait Islander people as the Traditional Custodians of the land.

EMV also acknowledges and pays respect to the Elders, past and present, and is committed to working with Aboriginal and Torres Strait Islander communities to achieve a shared vision of safer and more resilient communities.

Documentation

This document is available in Word and PDF format at www.emv.vic.gov.au/responsibilities /state-emergency-management-plan-subplans

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Version control

Title	Version date	Amendment
State Bushfire Plan 2012	June 2012	First version was based on provisions of the *Emergency Management Act 1986* and the *Fire Services Commissioner Act 2010*
State Bushfire Plan (Version 2 – 2014)	October 2014	An update of the first version to align the document with the *Emergency Management Act 2013* (which repealed the *Fire Services Commissioner Act 2010*) and the updated State Emergency Response Plan (July 2014 version)
State Emergency Management Plan Bushfire Sub-Plan	September 2021	The previous State Bushfire Plan was a sub-plan of the State Emergency Response Plan. This Sub-Plan is now updated to reflect the State Emergency Management Plan, published in September 2020 under reformed planning arrangements adapted into the *Emergency Management Act 2013* by the *Emergency Management Legislation Amendment Act 2018*.
State Emergency Management Plan Bushfire Sub-Plan	October 2023	An update to incorporate the implementation of the AFDRS, which commenced on 1 September 2022 and Victorian “Machinery of Government” (MoG) changes which took effect on 1 January 2023.

Contents

1. Introduction 8

1.1 About the State Emergency Management Plan Bushfire Sub-Plan 8

1.1.1 Purpose and scope 8

1.1.2 Audience 8

1.1.3 How to read this document 8

1.1.4 Plan preparation, approval and review 9

1.1.5 Plan activation 9

1.1.6 Concurrent emergencies 9

1.2 Legislative Framework 9

1.3 State Emergency Management Plan (SEMP) 10

1.3.1 SEMP Sub-Plans 10

1.3.2 This Plan and other key documents 11

1.4 State Emergency Management Priorities and Bushfire 13

1.4.1 Guiding principles 13

1.5 Victoria's Bushfire Threat 14

1.6 Shared Responsibility 15

1.6.1 Interoperability of fire agencies 18

1.7 Risk Management Standards 18

1.7.1 Strategic Bushfire Management Planning 19

1.7.2 Victorian Fire Risk Register – Bushfire (VFRR–B) 19

1.7.3 Bushfire Simulation Software 19

1.8 Evaluation and continuous improvement 20

1.8.1 Lessons management principles 20

1.8.2 Identifying lessons 21

1.8.3 Research 21

2. Mitigation and Prevention 22
2.1 Office of Bushfire Risk Management 22
2.2 Fuel management 22
2.2.1 Roadsides 23
2.2.2 Public Land 23
2.2.3 Private Land 24
2.3 Land use planning and building controls 25
2.4 Bushfire Prone Areas (BPAs) 25
2.5 Bushfire Management Overlay (BMO) 25
2.6 Powerline Safety 25
2.7 Fire Restrictions 26
2.8 Total fire ban days 26
2.9 Restricted access to public land 27
2.10 Arson and enforcement 27
2.11 Community education, awareness and engagement to prevent and respond to bushfire 27
3. Planning 28
3.1 State, Regional and Municipal Fire Management Planning 28
3.2 Strategic Bushfire Management Planning 29
3.3 Forecasting risk 30
3.4 Local Mutual Aid Plans (LMAPs) 30
3.5 Public media campaigns 31
4. Preparedness 32
4.1 Community preparedness 32
4.2 Community resilience 33
4.3 Neighbourhood Safer Places–Bushfire Place of Last Resort (NSP-BPLR) 34
4.4 Community Fire Refuges 34
4.5 Community information 34
4.6 Business preparedness 35
4.7 Household and individual preparedness 35
4.8 Australian Fire Danger Rating System (AFDRS) 35
4.8.1 Fire Danger Ratings 35
4.8.2 Fire Behaviour Index 36
4.8.3 Components of the Operational Build 37
4.9 Control Agency Preparedness 38
4.10 Human resources 38
4.11 Cross border arrangements 39

4.12 Capability 39
4.12.1 Training, Equipment and Information Communication Technology (ICT) 39
4.12.2 Shared ICT platforms 40
4.12.3 Aviation 40
4.12.4 Logistics 41
4.13 Intelligence and Predictive Services 41
4.13.1 Bureau of Meteorology 41
4.13.2 Intelligence products and Fire Behaviour Analysts (FBANs) 41
4.13.3 Remote sensing and other technology 41
5. Response 42
5.1 Concept of Emergency Operations 42
5.2 Tiers of Emergency Management 42
5.3 Readiness 43
5.3.1 Daily readiness arrangements 43
5.3.2 Resource readiness 43
5.3.3 Operational planning 44
5.4 Initial Response 44
5.4.1 Initial attack 44
5.4.2 Pre-determined dispatch of aircraft 45
5.4.3 Operational Safety 45
5.4.4 Safe Person Approach (SPA) 45
5.4.5 Dynamic Risk Assessment (DRA) 45
5.4.6 Red Flag Warnings 46
5.4.7 Field Information Update (FIU) 46
5.4.8 LACES 46
5.4.9 Health Monitoring 47
5.4.10 Hazard Trees 47
5.4.11 WATCHOUT 47
5.4.12 Occupational Health and Safety (OH&S) Incident Reporting 48
5.5 Control 49
5.5.1 Control Agency and Support Agencies 49
5.5.2 Determining the Control Agency 49
5.5.3 Transfer of Control 50
5.6 Coordination 51
5.7 Command 51
5.8 Incident Management 52

5.9 Incident levels 53
5.9.1 Level 1 incident 53
5.9.2 Level 2 incident 54
5.9.3 Level 3 incident 56
5.10 Incident Controller (IC) 56
5.11 Incident Management Team (IMT) 56
5.11.1 Deputy Incident Controller (Deputy IC) 57
5.12 Supplementary resources 57
5.12.1 Resource Requesting 57
5.12.2 Interstate and International Resources 57
5.13 Emergency Management Team(s) (EMT) 58
5.13.1 Incident Emergency Management Team (IEMT) 58
5.14 Control facilities 58
5.14.1 State Control Centre (SCC) 58
5.14.2 Regional Control Centres (RCCs) 59
5.14.3 Incident Control Centres (ICCs) 59
5.14.4 Control Point 59
5.14.5 Operations Point 59
5.14.6 Division and Sector Command Point 59
5.15 Operational communications 60
5.16 Bushfire naming convention 60
5.17 Traffic management 60
5.18 Evacuation 61
5.19 Smoke 62
5.20 Public Information 63
5.20.1 Provision of warnings to the community 63
5.20.2 Warning levels and templates 63
5.20.3 Levels of community warnings and information 63
5.20.4 All Clear Messages 64
5.20.5 Community Information 64
5.20.6 Publishing Warning and Information Areas (Polygon) 64
5.20.7 Community meetings 65
5.20.8 Warning channels 65
5.20.9 Community Alert Sirens 65

5.21 Relief (as a part of response) 66
5.21.1 Initial Impact Assessment 67
5.22 Animal (livestock and companion) and wildlife welfare 67
5.23 Consequence Management 67
5.24 Transition to Recovery 68
6. Recovery 69
6.1 Public land recovery 70
6.2 Fences and control lines after bushfires 70
6.3 Essential water replacement after bushfire 71
7. Document information 72
7.1 Acronyms and Terms 72
7.2 Linked resources 76

1. Introduction

1.1 About the State Emergency Management Plan Bushfire Sub-Plan

1.1.1 Purpose and scope

Victoria's State-level emergency risk assessment, Emergency Risks in Victoria, was published in 2020 and set Victoria's emergency risk profile. Bushfire was assessed as a 'State significant risk', meaning bushfire was assessed as being a critical and credible scenario for the State.

This State Emergency Management Plan (SEMP) Bushfire Sub-Plan (this plan) outlines the current arrangements for the management of bushfires in Victoria. This supports a comprehensive, integrated and coordinated approach and reflects a shared responsibility for bushfire management. In alignment with the SEMP, this plan contextualises the current arrangements, roles and responsibilities for bushfire mitigation, planning, preparedness, response (including relief) and recovery.

The Department of Energy, Environment and Climate Action (DEECA) coordinated the 2023 update of this plan in conjunction with the other fire services agencies: Country Fire Authority (CFA), and Fire Rescue Victoria (FRV), with input from a range of other emergency management agencies.

This plan does not include arrangements for the management of structural fires.

1.1.2 Audience

This plan recognises that emergency management and supporting communities to be safer and more resilient is the shared responsibility of all Victorians, not just the emergency management sector.

The audience for this plan comprises government, business, the community of Victoria and, primarily, agencies within the emergency management sector.

1.1.3 How to read this document

This plan should be read in conjunction with the SEMP.

Acronyms

A list of acronyms used throughout this plan is included at section 7.1 and is consistent with the acronyms used throughout the SEMP.

Hyperlinks

This plan refers to a range of existing resources relating to bushfire, including documents and websites. This plan does not seek to duplicate the information contained in these resources and instead provides links to where the reader can obtain further information.

For more operational or sensitive information, a log-in may be required, such as for documents saved on the Emergency Management Common Operating Picture (EM-COP; see section 4.12.2), including Joint Standard Operating Procedures (JSOPs).

Documents or resources that are referred to frequently throughout this plan (such as the SEMP) are not hyperlinked in each instance however a table outlining all links provided throughout this plan is included at section 0 for completeness.

All hyperlinks were accurate at time of publication.

1.1.4 Plan preparation, approval and review

This is the fourth version of a State-tier bushfire plan, and the second in the form of a SEMP Bushfire Sub-Plan that is prepared under the reformed emergency management (EM) planning arrangements outlined in the *Emergency Management Act 2013* (EM Act 2013), as amended by the *Emergency Management Legislation Amendment Act 2018.*

This plan aligns with the SEMP and was prepared with regard to the Guidelines for Preparing State, Regional and Municipal Emergency Management Plans; and was approved by the State Crisis and Resilience Council (SCRC) on 29 November 2023. This plan was published and took effect on December 2023.

The EM Act 2013 requires this plan to be tested, reviewed and updated at least every three years with consideration given to earlier revisions as required to ensure the plan provides for a current, integrated, coordinated and comprehensive approach to bushfire management at a State level. Earlier reviews may be triggered by the sub-plan being applied in a major emergency or exercise or following a substantial change to relevant legislation or arrangements, including the SEMP.

This plan was reviewed in 2023 due to sector changes in emergency management relevant to the management of bushfires, as well as the implementation of the Australian Fire Danger Rating System (AFDRS).

1.1.5 Plan activation

The arrangements in this plan apply on a continuing basis and do not require activation.

1.1.6 Concurrent emergencies

In the case of a concurrent emergency (e.g. extreme heat or an energy disruption), the arrangements detailed in this plan may need to be adjusted as required. To affect this the State Response Controller – Class 1 should consult with the Emergency Management Commissioner (EMC) and other relevant State Controllers, to determine (and document) appropriate control arrangements.

1.2 Legislative Framework

The following legislation, while not exhaustive, is the principal legislation for bushfire management in Victoria:

- *Emergency Management Act 2013* (EM Act 2013)
- *Emergency Management Act 1986*
- *Country Fire Authority Act 1958* (CFA Act 1958)
- *Fire Rescue Victoria Act 1958* (FRV Act 1958; formerly the *Metropolitan Fire Brigades Act 1958*)
- *Forests Act 1958*.

In addition to these, the Code of Practice for Bushfire Management on Public Land is a legislative instrument established under the *Conservation, Forests and Lands Act 1987* that defines the objectives, principles, and approach to the management of bushfires on public land in Victoria.

1.3 State Emergency Management Plan (SEMP)

Under the EM Act 2013, the EMC is required to arrange for the preparation of the SEMP. The SEMP provides for an integrated, coordinated, and comprehensive approach to emergency management at the State level. It contains provisions for the mitigation of, response to and recovery from emergencies (before, during and after), and specifies the roles and responsibilities of agencies in relation to emergency management.

The mitigation, response (including relief) and recovery arrangements for emergencies are outlined in the SEMP, including the role of the EMC.

The EM Act 2013 defines a ***major fire*** as a large or complex fire (however caused) that:

- has the potential to cause, or is causing, loss of life and extensive damage to property, infrastructure or the environment or
- has the potential to have, or is having, significant adverse consequences for the Victorian community or a part of the Victorian community or
- requires the involvement of two or more fire service agencies to suppress the fire or
- will, if not suppressed, burn for more than one day.

Under the EM Act 2013, major fires are a Class 1 emergency, given a Class 1 emergency is either a major fire or any other major emergency for which CFA, FRV or the Victoria State Emergency Service Authority (VICSES) is the control agency.

1.3.1 SEMP Sub-Plans

The Guidelines for Preparing State, Regional and Municipal Emergency Management Plans provide more information about the requirements relating to SEMP sub-plans; they are developed to define more complex or specific arrangements than those contained in the SEMP. This plan seeks to add further detail about the specific arrangements for the management of bushfires in Victoria than what is outlined in the SEMP.

1.3.2 This Plan and other key documents

The below table indicates key documents for the State's bushfire context. This is not an exhaustive list. For example, agencies with a role in bushfire management may develop agency-specific plans to manage specific risks that are not included in this table. Likewise, doctrine and agency Standard Operating Procedures (SOPs) apply at all tiers and so are not represented in the table.

Bushfires have far-reaching impacts for people, communities and the environment. In some cases, this may require the activation of other emergency management arrangements to manage significant consequences and other emergencies caused by bushfire (such as energy supply disruptions or impacts to wildlife).

Table 1: Plans and documents for the State's bushfire context

	Community resilience	Emergency Management Plans	Operational arrangements, and agency-specific plans
National	National Disaster Risk Reduction Framework National Bushfire Management Policy Statement for Forests and Rangelands Strategic Directions for Fire and Emergency Services in Australia and New Zealand 2022-2026		Agreements for resource sharing Australian Disaster Resilience Handbook Collection
State	Strategic Roadmap for Emergency Management in Victoria 2022–28 Safer Together Policy Bushfire Safety Policy Framework Victorian Planning Provisions	State Emergency Management Plan **SEMP Bushfire Sub-Plan (this Plan)**	Code of Practice for Bushfire Management on Public Land Joint agency SOPs Joint agency Business Rules Interagency MOUs
Regional	Victorian planning schemes	Regional Emergency Management Plans (REMPs) Optional: Regional Fire Management Sub-Plans (sub-plans of REMPs)	CFA/DEECA/FRV/SES Local Mutual Aid Plans CFA Regional Operational Management Plans CFA/DEECA Regional Bushfire Management Strategies CFA/DEECA) Joint Fuel Management Program

	Community resilience	Emergency Management Plans	Operational arrangements, and agency-specific plans
Municipal	Resilient Recovery Strategy	Municipal Emergency Management Plans (MEMPs) Optional: Municipal Fire Management Sub-Plans (sub-plans of MEMPs)	
Community	Community Resilience Framework for Emergency Management	Optional: Community led action plans	Pre-incident Response Plans
Household		Household Plans	

1.4 State Emergency Management Priorities and Bushfire

The State Emergency Management Priorities are outlined in the SEMP and underpin and guide all decisions during a response to any emergency, including bushfires.

The Incident Controller (IC) may need to vary the priorities in some circumstances. This shall be recorded, reported and disseminated in consultation with the Regional Controller (RC) and State Response Controller (SRC) based on sound incident predictions and risk assessment.

1.4.1 Guiding principles

The following guiding principles are supplementary to the State Emergency Management Priorities.

Leadership

The EMC and the fire services agencies are accountable, on behalf of the Victorian Government, for leading other agencies, the community and individuals to make appropriate arrangements to reduce the impact and consequence of bushfire.

Protection of human life

As per the State Emergency Management Priorities, the protection and preservation of human life and relief of suffering, which includes the lives of both community members and emergency services personnel, takes priority above all other considerations in bushfire management.

Responsibility for building resilience

Bushfires are inevitable and not all bushfires are preventable. All levels and sectors of society share responsibility, within their sphere of influence, for building a more resilient community and environment that can prevent, respond to and recover from bushfire.

Community involvement

Community involvement is essential to ensure bushfire management approaches are inclusive, integrated, and comprehensive across diverse communities and landscapes.

A seamless approach

The fire agencies work together with emergency management sector partners, using resources efficiently and effectively, to provide the community with a seamless approach to all aspects of bushfire management.

Risk-driven

Plans and priorities for bushfire management should aim to achieve a measurable reduction in the impact and consequences of bushfire.

Learning and knowledge

Local knowledge, experience, and operational and scientific evidence are all integral to the ongoing improvement of bushfire management policy and practice.

1.5 Victoria's Bushfire Threat

Victoria is one of the most bushfire-prone areas in the world and has had regular bushfires since European records began in the mid-1800s. The last two decades have seen a dramatic increase in the number, size, extent and severity of bushfires. Prior to 2002-03, the previous century had recorded one fire season (1938-39) where more than one million hectares of Victoria were burnt by bushfires; there were three such seasons between 2002-03 and 2019-20 (2002-03, 2006-07 and 2019-20). Some of the State's most catastrophic fires include Black Thursday in 1851, Black Friday in 1939, Ash Wednesday in 1983 and Black Saturday in 2009 and most recently Black Summer in 2019–20.

The 2019-20 fires provided further evidence that climate change is increasing the frequency and severity of seasonal bushfire conditions throughout Australia's east, extending from Victoria to Queensland. 2019 was Australia's hottest and driest year recorded. Record heat and drought leading up to the summer of 2019-20 primed much of eastern Australia for bushfires and, by the end of January 2020, approximately 21 per cent of all of Australia's forests had burnt.

In Victoria, the fire season is now starting at least a month earlier than it was in the 1990s and an increase in the number of days with an elevated fire behaviour index is having an impact on the fire activity we are experiencing. We have seen an increase in the number of fires and in the last 20 years we have had four fire seasons (2002-03, 2005-06, 2008-09 and 2019-20) that each had over 400,000 hectares burnt, all stretching our capacity to respond and having devastating consequences for the community and environment.

Models of future climate show that this pattern is likely to continue – conditions will be hotter, drier and with more extreme weather events. This is expected to lead to worse fire seasons, including the potential for more frequent large fires (like the 2019-20 season) and more extreme fire behaviour (including an increase in fire generated thunderstorms). This will have flow on effects to the safety of the community and firefighters.

Bushfire is a significant risk facing the State and is also a natural part of the Victorian environment that drives regeneration and maintains the health of some species and ecosystems.

Bushfire can be ignited by natural causes such as lightning, or human activity. The high bushfire risk is the consequence of a combination of factors including:

- large areas of highly flammable dry eucalypt forest
- expanses of highly flammable grassland
- a climatic pattern of mild, moist winters followed by hot dry summers
- protracted droughts
- agricultural practices where fire is used routinely
- an increasing population density in bushfire-prone areas, such as in the peri-urban fringe.

1.6 Shared Responsibility

Royal Commissions and inquiries continue to highlight how critical a shared responsibility approach is to emergency management. Victoria's approach to bushfire management is underpinned by the principle of shared responsibility and involves all levels of government, community, individuals and the private sector. Integration and collaboration are at the core of existing Victorian Government arrangements for bushfire management.

The SEMP recognises that developing safer and more resilient communities is a shared responsibility of all Victorians, not just the emergency management sector. That said, Victoria's fire services agencies hold the primary responsibility for bushfire management in their respective jurisdictions. The State has a duty to prevent, plan and prepare for, respond to and support recovery from bushfires.

In Emergency Management, the concept of 'shared responsibility' refers to the collective obligations and accountabilities held by a range of actors. A commitment to shared responsibility recognises that no single actor can be responsible for emergency mitigation, preparedness, response or recovery, and that shared responsibility in emergency management is everyone's business. Individuals, communities, organisations, businesses, all levels of government and the not-for-profit sector all have some role to play in planning for, responding to and recovering from emergencies.

Government, across a wide range of ministerial portfolios, develops legislation, and plans and implements programs to reduce the risk of bushfire. Although the protection and preservation of life and relief of suffering is paramount, during a bushfire, the State cannot absolutely guarantee the safety of all community members.

Agencies are encouraged to seek genuine participation from the community, empowering them to achieve shared responsibility.

All roles and responsibilities for the mitigation of, preparedness for, response to and recovery from bushfire are as per the SEMP:

- The SEMP outlines participating agencies for mitigation activities specific to bushfire
- The SEMP identifies three control agencies for bushfire response: CFA, DEECA and FRV.
- The Chief Officer (CFA)/Chief Fire Officer (DEECA)/Commissioner (FRV) of each fire services agency is a statutory officer, accountable for delivering the bushfire-related responsibilities of their respective agency.

Table 2: Actors and functions of responsibility

Actor	Agency	Function
Emergency Management Sector	CFA	Responsible for the prevention and suppression of fire in the country area of Victoria (private property outside the FRV fire district). Responsible for leading community education and preparedness. Detailed role and responsibilities are outlined in the SEMP's CFA role statement.
	DEECA	Responsible for the prevention and suppression of bushfire in State forests, national parks and on protected public land. Responsible for coordinating recovery of the natural environment, public land and waterways following bushfire. Following the 2019-20 bushfires, the Victorian Government established the Office of Bushfire Risk Management within DEECA responsible for overseeing the development and implementation of an end-to-end framework for bushfire risk management, with a primary focus on fuel management. The Office's role does not extend to seasonal preparedness, readiness and response activities, or the on-ground delivery of bushfire risk reduction activities. Statutory authorities remain accountable for their statutory functions. Land managers and landholders retain primary responsibility for delivering bushfire risk reduction activities on their land. Detailed role and responsibilities are outlined in the SEMP's DEECA role statement.
	FRV	Responsible for the prevention and suppression of fire in the FRV fire district. Detailed role and responsibilities are outlined in the SEMP's FRV role statement
	Department of Transport and Planning (DTP)	Responsible for developing policy and regulation to mitigate bushfire through land use planning and building. Responsible for the protection and management of listed historic heritage places through mitigation of and recovery from bushfire. Responsible for taking all practicable steps to prevent the occurrence of fires and minimise the danger of the spread of fires on and from any land or roads under DTP control or management (as per the *CFA Act 1958*). Responsible for providing fire agencies with access (via permits) to Department owned land to conduct fuel management activities as per *Road Management Act 2004*. Responsible for reopening of transport access to communities (including to support restoration of power and telecommunications) once it is safe to do so.
	Response Support Agencies (RSAs)	See the SEMP table: Response support agencies, by functional area for the lead response support agency.

Actor	Agency	Function
	Councils	Responsible for administering and enforcing the planning scheme, including ensuring strategic settlement planning activities and development responds to bushfire hazard and risk. Responsible for appointing a Municipal Fire Prevention Officer (MFPO), Municipal Emergency Management Officer (MEMO) and Municipal Recovery Manager (MRM). A significant focus is on land use planning, Responsible for hazard mitigation on council-owned or managed land, fire prevention programs on private land (including the issuing of fire prevention notices and permits), broader emergency planning and response support. Responsible for taking all practicable steps to prevent the occurrence of fires and minimise the danger of the spread of fires on and from any land or roads under council control or management (as per the *CFA Act 1958*). Councils play a key role in designating and recording Neighbourhood Safer Places-Bushfire Places of Last Resort (NSP-BPLRs) as last resort shelter options during bushfires (see section 0). During and following bushfires, councils are responsible for coordinating the relief and recovery services and support for local communities. This includes establishing Emergency Relief Centres and conducting secondary impact assessments. Detailed role and responsibilities are outlined in the SEMP's Municipal Council role statement.
Individuals and households		Individual community members are responsible for learning the risks of their environment and making decisions about how to respond to risks commensurate with their risk appetite. Owners and occupiers of land are responsible to manage their properties to reduce the risk of bushfire (see section 0). Each household is responsible for preparing their property for bushfire for example, retrofitting existing dwellings and clearing vegetation around the house, in accordance with State planning controls for the clearing of native vegetation and planning scheme requirements. Information to assist householders and landowners to prepare their property for bushfire can be found on the CFA website. Every household should have a well-practised written bushfire survival plan.
Small, medium and large businesses		Every business and farm should have a well-practised written bushfire survival plan.
Community groups and networks		As per the SEMP.

1.6.1 Interoperability of fire agencies

Interoperability provides a mechanism for achieving better outcomes by allowing the control and support agencies to effectively work together before, during and after an incident. It also provides a foundation for meaningful connections with the community and a wide range of partner agencies (see section 5.5.1. for information about control and support agencies).

To achieve a cooperative response to bushfire emergencies, each fire agency maintains an understanding of the systems, structures, resources, capabilities and statutory obligations of the other agencies, and there are also multiple inter-agency agreements to allow operations across fire districts and define how agencies interact in those circumstances. Interoperability maximises the capability of the agencies to work effectively and efficiently together to deliver information, communications, and technology.

The Australasian Interservice Incident Management System (AIIMS) provides the basis for control structures. Joint Standard Operating Procedures (JSOPs) for bushfire management are developed and agreed between relevant agencies (such as fire agencies) and are managed by EMV.

Fire agencies also work collaboratively with a wide range of industry bodies, community groups and other stakeholders regarding bushfire-related issues. This includes collaboration with Victorian, interstate and international fire and emergency management agencies in order to share knowledge, improve bushfire and emergency management policy and practice and to optimise the use of resources.

1.7 Risk Management Standards

The adoption of emergency risk management, and its underlying concept of risk, has helped communities and emergency managers move beyond the narrower concept of ‘hazard’ to ‘risk’. Risk deals with the interaction of the exposure to hazard and specific vulnerabilities.

Bushfire has been assessed as a significant risk for Victoria in all State-level multi-hazard emergency risk assessment projects, including studies undertaken in 2008, 2012 and 2017. The results of the 2017 assessment are published in Emergency Risks in Victoria (July 2020). The report is scheduled to be updated following review and approval in November 2023.

Approaches for managing bushfire risk in Victoria include the application of customised risk management systems, planning frameworks and tools, tailored for reducing bushfire risk. This includes:

- Bushfire Risk Engagement Areas (BREAs; see section 0)
- Fire Management Zones (FMZ)
- Strategic bushfire management planning (see section 0)
- Victorian Fire Risk Register – Bushfire (VFRR–B; see section 1.7.2) and
- Bushfire Simulation Software (see section **Error! Reference source not found.**).

1.7.1 Strategic Bushfire Management Planning

Managing bushfire risk is an ongoing and shared responsibility. Local communities are now involved in decision making about bushfire management all year round and determine local solutions to reduce bushfire risk.

Victoria has adopted a risk-based approach to bushfire management. Bushfire risk reduction targets and decision support tools inform the timing and location of fuel management activities, ensuring they are prioritised for delivery where they deliver the greatest impact in reducing risk.

The fire agencies have undertaken to work with communities to develop regional bushfire management strategies through a strategic bushfire management planning process that:

- identify values to be protected from bushfire;
- assesses bushfire risk to those values; and
- set out strategies to manage this risk.

1.7.2 Victorian Fire Risk Register – Bushfire (VFRR–B)

VFRR-B is a process in which representatives from council, fire services, public land managers, utilities and community groups map assets at risk from bushfire and assess the level of risk to the assets. Assets may include residential areas, children's services, hospitals, aged-care facilities, infrastructure, commercial industry, tourism events, and those that are culturally significant. Agency representatives also record the current treatments which are carried out to mitigate the risk to the asset.

The VFRR-B can support the preparation of Municipal Emergency Management Plans (MEMPs) or MEMP sub-plans (see section 3.1). Several agencies also use VFRR-B as an evidence-based data to support bushfire planning and decision-making.

1.7.3 Bushfire Simulation Software

Phoenix RapidFire (Phoenix) is a sophisticated bushfire simulation tool used to model fuel-driven bushfire risk. Phoenix uses information about weather, topography, vegetation and fire history to simulate and predict the spread and impact of bushfires. It helps to understand bushfire behaviour – including flame height, ember density, spotting distance, convection column strength and intensity.

CFA and DEECA, use Phoenix to understand how different fuel management activities affect the amount and location of bushfire risk across landscapes. Phoenix is one of the tools used to help identify the most effective fuel management strategies to reduce bushfire risk.

Phoenix is also used to support bushfire preparedness, informing the preparation and positioning of firefighters, aircraft and equipment on days of elevated fire danger, as well as in response, including planning attack strategies, evacuations and potential impacts.

Phoenix bushfire simulations are also used to help communities understand how bushfires behave. Maps and simulations are used in conversations with, and advice to, communities about what bushfire risk means for them in both the readiness and response phases of bushfire management.

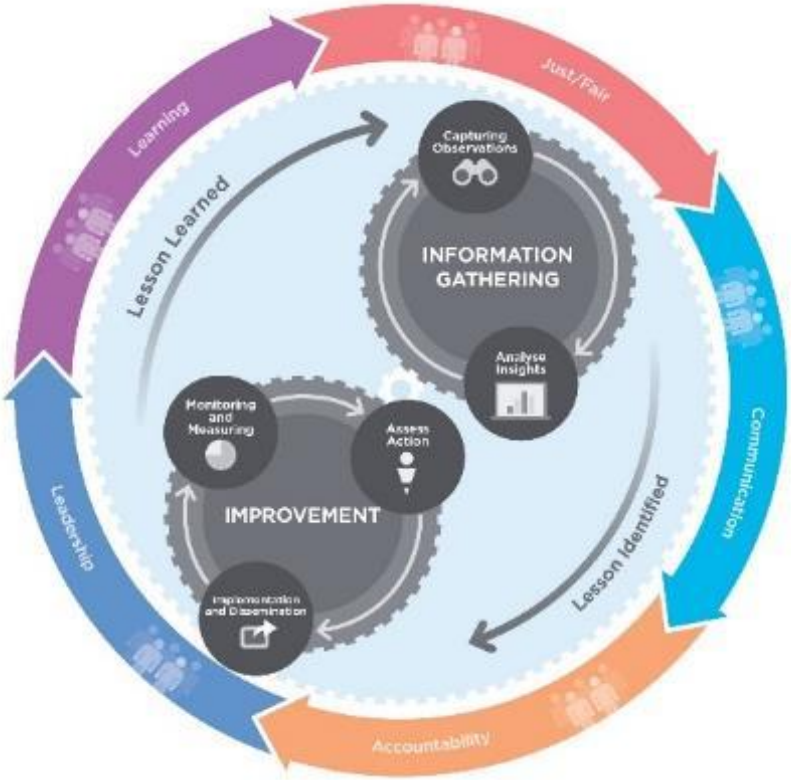
Victorian fire agencies continue to review current and emerging bushfire simulation software to ensure that bushfire modelling is as accurate as possible. In February 2021, a partnership between CSIRO's Data61 and the Australian and New Zealand National Council for fire and emergency services (AFAC) announced the development of a nationally consistent bushfire modelling and prediction capability, known as Spark. Agencies will review how this simulation software could be incorporated into the current risk modelling approach and operational predictions used in Victoria.

1.8 Evaluation and continuous improvement

1.8.1 Lessons management principles

The emergency management sector's approach to identifying and learning operational lessons, including for bushfire operations is documented in EM-LEARN Lessons Management Framework and includes the lessons management life cycle below.

Figure 1: Victorian's Lessons Management Life Cycle



Source: EM-LEARN Lessons Management Framework

1.8.2 Identifying lessons

The process of identifying lessons is based on a trend analysis of observations collected through learning activities, including exercising, monitoring, debriefing and reviewing. An observation is a record of a noteworthy fact or occurrence that someone has heard, seen, noticed or experienced as an opportunity for improvement or an example of good practice.

Observations are collected through various processes, including:

- Exercises (refer to Australian Institute for Disaster Resilience (AIDR) Managing exercises handbook)
- Real time monitoring and evaluation (refer to JSOP012.01 – Real Time Monitoring and Evaluation)
- Operational debriefing, including hot debriefs, informal debriefs, seasonal debriefs or After-Action Reviews
- Operational reviews, including agency-specific, multi-agency or external reviews
- Directly through online observations capture tools, including surveys and EM-Share.

Operational learning activities are the responsibility of the relevant agency or emergency management tier and should be conducted in line with any relevant guidelines, policy and legislation. It is good practice to invite representatives of relevant community, business and industry groups to debriefs where possible. Often, a post-incident community forum may be appropriate.

1.8.3 Research

Over the past century, Victoria has amassed a wealth of science-based knowledge about bushfire behaviour and management that has fundamentally changed the way bushfire is managed across the State. This science is nationally and internationally significant and has led to Victorian communities being safer.

The fire services agencies are involved in several ongoing research programs in partnership with a range of other organisations, such as Natural Hazards Research Australia (NHRA; which has transitioned from the Bushfire Cooperative Research Centre (Bushfire CRC) and the Bushfire and Natural Hazards Cooperative Research Centre (BNHCRC)), universities and industry.

NHRA is Australia's national centre for natural hazard resilience and disaster risk reduction. Its mission is to work with partners (which includes state and local government) and the community on research that is useful, actionable and supportive of better decision-making to save lives and protect communities.

NHRA comprises the fire and land management agencies in Australia and New Zealand, universities, state and territory governments, the Bureau of Meteorology, the Australian Government Attorney General's Department and several other fire and emergency management related agencies.

2. Mitigation and Prevention

Emergency Risks in Victoria (July 2020) outlines broad mitigation strategies for bushfire. This document is due for re-publishing in November 2023.

Managing bushfire risk is an ongoing and shared responsibility and everyone has a role. Victoria is moving to a more integrated approach across public and private land, with fuel management just one of a range of different management actions we take to protect lives, homes, jobs and the environment.

2.1 Office of Bushfire Risk Management

The Victorian Government has established the Office of Bushfire Risk Management (OBRM). OBRM sits within DEECA and is supported by an advisory panel.

The OBRM is responsible for overseeing the development and implementation of an end-to-end framework for bushfire risk management, with a primary focus on fuel management. The Office's role does not extend to seasonal preparedness, readiness and response activities, or the on-ground delivery of bushfire risk reduction activities. Statutory authorities remain accountable for their statutory functions. Land managers and landholders retain primary responsibility for delivering bushfire risk reduction activities on their land.

2.2 Fuel management

The Joint Fuel Management Program (JFMP) is a State-wide program that manages fuel on public and private land over a rolling three-year period. The JFMP is updated annually with the latest information, and built on the current long term regional Bushfire Management Strategies.

The JFMP prescribes a range of delivery activities to reduce the amount of fuel and fuel-driven bushfire risk in a landscape. Methods used include planned burning, and mechanical works such as slashing, mowing and clearing vegetation. Other operational programs inform the creation and maintenance of strategic fuel breaks and carrying out fire infrastructure maintenance (such as roads and bridges, fire dams and lookout towers).

Fuel management helps reduces the spread, intensity and severity of bushfires across the landscape. This contributes to reducing impacts on communities and the environment.

Fuels are managed differently across different areas in Victoria. In areas close to towns, fuel treatments are more frequent to protect people and the things they value.

In other areas, burns can be managed to reduce the spread of bushfires across the landscape or promote ecological outcomes. Some areas are excluded from planned burning to protect areas that are less tolerant of fire.

CFA, Forest Fire Management Victoria (FFMVic) and FRV partner with Traditional Owner Corporations and groups to support implementation of their cultural fire priorities as set out in, but not limited to, Whole of Country plans and the Victorian Traditional Owner Cultural Fire Strategy (VTOCS).

In high risk areas, mechanical treatment is carried out as either an alternative treatment option to planned burning in whole or part, particularly where treatment provides immediate bushfire risk reduction or assists delivery of a burn at a future time.

CFA, FFMVic and FRV firefighters routinely work together on planned burns.

2.2.1 Roadsides

Under the CFA Act 1958 and FRV Act 1958 it is the responsibility of every municipality and public authority (including the Department of Transport and Planning, DEECA and Parks Victoria) to take all practicable steps, including burning, to prevent the occurrence of fires and minimise the danger of the spread of fires on and from any land under its control and any road under its care and management.

Councils and road managers routinely work with fire agencies through an integrated planning process to identify, plan and deliver a planned program of works that aims to mitigate the risks associated with fire and roadside fuels. Typical roadside fuel management treatments include mowing, slashing, spraying and burning.

2.2.2 Public Land

Approximately one third of Victoria is public land managed by DEECA, Parks Victoria and other public authorities. The Code of Practice for Bushfire Management on Public Land provides the primary objectives for bushfire management on public land in Victoria which are:

- to minimise the impact of major bushfires on human life, communities, essential and community infrastructure, industries, the economy and the environment. Human life will be afforded priority over all other considerations.
- to maintain or improve the resilience of natural ecosystems and their ability to deliver services such as biodiversity, water, carbon storage and forest products.

To support fire and emergency management, DEECA) maintains a network of approximately 50,000 kilometres of fire access roads.

Fuel management on public land is informed by the four Fire Management Zones (FMZs):

- **Asset Protection Zone** - An area around properties and infrastructure where fuel is managed intensively to provide localised protection to reduce radiant heat and ember attack on life and property in the event of a bushfire.
- **Bushfire Moderation Zone** - An area around properties and infrastructure where fuel is managed to reduce the speed and intensity of bushfires and to protect nearby assets, particularly from ember attack in the event of a bushfire.
- **Landscape Management Zone** - An area where fuel is managed to minimise the impact of major bushfires, to improve ecosystem resilience and for other purposes (such as to regenerate forests and protect water catchments).
- **Planned Burning Exclusion Zone** - An area where planned burning is avoided, mainly because ecological assets in this zone are less tolerant of fire.

2.2.3 Private Land

Private landowners and occupiers are responsible for ensuring their activities, use of fire and the level of vegetation on their land does not present an unacceptable danger to life or property from the threat of fire.

As part of the strategic bushfire management planning process, land and fire management agencies have undertaken an analysis to define Bushfire Risk Engagement Areas (BREAs). BREAs identify parts of the landscape where managing bushfire fuels is most effective in reducing risk. This helps to indicate the priority areas in a region where agencies can work with communities to reduce bushfire fuels.

BREAs help focus conversations on the range of treatment options to reduce bushfire risk. This may include other actions where reducing fuels may not be possible. On-ground discussions and assessments between agencies and the community will determine the treatments that best suit a particular place. Other measures for preparing a property include retrofitting existing dwellings so they can better withstand bushfires.

Landowners and occupiers are encouraged to remove fire hazards and manage fuels around their property before the Fire Danger Period. This routinely includes cutting long grass, removing fallen timber and rubbish, and clearing leaves and branches from gutters. Information on the clearing of native vegetation around houses can be found by contacting the relevant council.

To assist property owners in the management of vegetation on their land and mitigate their bushfire risk, specific planning permit exemptions exist in Victorian planning schemes to allow the clearance of vegetation around certain dwellings to create or maintain defendable space that assists bushfire protection. The level of hazard reduction on private land can influence fire behaviour, survivability and can either complement or detract from the effectiveness of hazard reduction on adjoining private and public land. Where unacceptable risks are identified, Municipal Fire Prevention Officers (MFPOs) appointed under the CFA Act (1958) and the FRV Act (1958) enforce fuel hazard management on private land. Authorised Officers under the Forests Act 1958 may also direct any owner or occupier or person or body of persons having the control or management of any land within 1·5 kilometres (unless excised) of the boundary of any State forest, national park or protected public land - to remove, destroy by burning or abate in any manner directed, a fire hazard on such land.

Farmers, farm managers and other primary produces have special need to consider fire safety, actively plan for and mitigate fire risk, manage vegetation and on occasion undertake farming activities that present a fire risk. Fire safety on the farm | CFA (Country Fire Authority) content has been developed for farmers and other rural landholders.

Planned burning is conducted by CFA brigades on behalf of and at the request of private landowners or managers of other reserves such as roads and rail corridors, council reserves and water authority land. Many of these landowners and managers are actively developing capabilities to support an expanded fuel management program.

2.3 Land use planning and building controls

Land use planning and building systems are key risk mitigation measures in Victoria by regulating land use and development to ensure bushfire hazard and risk are considered and managed. They influence where growth and development occur and ensure that development can better withstand impacts. The systems also help communities recover following an emergency by streamlining processes relating to temporary buildings for housing and business, and rebuilding.

Land use planning that considers natural hazard risks is a critical mitigation measure in preventing future disaster losses in areas of new development.

Working with councils and the fire services agencies, the Department of Transport and Planning (DTP) administers the land use planning and building systems relevant to bushfires. Land use planning and building systems are important for creating more bushfire resilient communities. Particularly, strategic planning, through local planning schemes is critical in setting out how settlements and rural areas will grow and change in response to the threat of bushfire. Building regulations ensure new buildings are constructed with regard to likely forms of bushfire attack.

Bushfire mapping is a key element of the land use planning and building regulatory framework. Mapping criteria identifies whether an area should be designated a Bushfire Prone Area (BPA; see section 2.4), and if a Bushfire Management Overlay (BMO; see section 2.42.5) should apply.

2.4 Bushfire Prone Areas (BPAs)

Victoria's BPAs are designated areas under the *Building Act 1993* where the bushfire hazard has been identified and mapped. They make up most of the State.

These areas are subject to or likely to be subject to bushfires. These areas are subject to building permit requirements, where new buildings are required to build to a national bushfire construction standard. This is known as a Bushfire Attack Level (BAL).

2.5 Bushfire Management Overlay (BMO)

The BMO is a planning control that applies to bushfire prone areas with very high and extreme bushfire hazards. These areas are subject to planning permit requirements including mandatory bushfire protection measures such as defendable space, water supply, access and ongoing vegetation management requirements. Areas where a BMO applies are also by default BPAs (see section 2.4). Bushfire may also need to be considered in planning proposals which are outside a BMO but within a BPA.

2.6 Powerline Safety

Electricity powerlines contacting vegetation is one cause of bushfire. The *Electricity Safety Act 1998* requires electricity distribution companies to prepare bushfire mitigation plans, which are audited by Energy Safe Victoria. The *Electricity Safety Act 1998* also creates the Electric Line Clearance Consultative Committee, which manages a code to ensure vegetation remains clear of powerlines.

Energy Safe Victoria works with power companies to install new protection mechanisms to reduce the risk of starting fires, and work closely with fire agencies to review expected conditions to assist in identifying priority areas over high-risk periods. The bushfire mitigation programs of electricity distribution companies typically include activities such as asset inspection, maintenance, construction, upgrading, replacement, vegetation management, performance monitoring and auditing.

2.7 Fire Restrictions

CFA and DEECA use fire restrictions to reduce the likelihood of bushfire ignition through regulating activities known to start fires on days of elevated fire danger or throughout the high-risk weather season. The fire services actively work with Victoria Police to enforce any breaches. During periods in which fire restrictions apply, permits may be issued by CFA, DEECA , and FRV in certain circumstances to individuals that provide for otherwise restricted activities to be undertaken within a defined set of prescriptions.

During periods of increased fire risk:

- CFA declares and terminates the Fire Danger Periods by municipalities (or part thereof) in the country area of Victoria across Fire Districts.
- DEECA declares and terminates the seasonal Prohibited Period across private property located within 1.5 kilometres of a State forest, national park, and protected public land situated within 14 municipalities and 6 Alpine Resorts in the east of Victoria. Because the 1.5km margin overlays the Country Area of Victoria, there are pockets of land where both DEECA's Prohibited Period or CFA's seasonal restrictions (the Fire Danger Period) may apply.

These declarations are published in the Victorian Government Gazette. The status of the declarations can be found on the CFA and FFMVic websites.

The Prohibited Period in State forests, national parks, and on protected public land remains in force all year round. The *Forests (Fire Protection) Regulations 2014* outline activities that can be undertaken without a permit, however, certain conditions must be followed.

Councils also have local laws restricting activity involving fire. Information on these can be obtained from the relevant council.

2.8 Total fire ban days

Section 40 of the *CFA Act 1958* provides for the CFA Chief Officer to declare a Total Fire Ban (TFB) on days when the danger of fires occurring is elevated where any fire that occurs will be extremely difficult to control. The declaration of a TFB imposes strict rules and regulations regarding the circumstances in which fires may be lit and activities which pose a risk of causing a fire must be conducted. TFBs are declared for the whole part of Victoria or parts thereof defined by the Bureau of Meteorology's weather districts.

CFA, DEECA and FRV may issue permits for their respective jurisdictions under section 40 of the *CFA Act 1958* on days of TFB to allow for activities such as essential catering, welding, grinding, gas flare off, hot air ballooning and heating and spreading of bitumen.

The penalties for lighting fires illegally on TFBs, during the Fire Danger Period and during the Seasonal/Prohibited Period include large fines and possible imprisonment.

2.9 Restricted access to public land

The *Safety on Public Land Act 2004* (section 4(1)) creates a provision to declare public safety zones in State forests.

Some of the intents of public safety zones are to:

- protect flora and fauna to allow rehabilitation of a site in a heavily used area.
- allow for forest regeneration following a bushfire or other disturbance.
- allow the orderly and safe management of fuel reduction and fire suppression activities.

On days of forecast Catastrophic Fire Danger Rating, DEECA and Parks Victoria will close parks and forests (including State forests,national parks, state parks and other reserves) in the bushfire prone areas of the relevant weather district for public safety. In certain cases, exemptions to closure when the forecast fire danger rating is catastrophic are approved by the Chief Executive Parks Victoria. These exemptions may include parks managed by Parks Victoria in urban areas, and marine and coastal parks.

2.10 Arson and enforcement

Victoria Police, in recognising the serious impact bushfire can have on communities, undertakes an intelligence-led, partnership approach to preventing the incidence of bushfire arson, preserving life, protecting property and detecting offenders.

Enforcement links inextricably to community education, because the main aim is to change the behaviour of people.

The fire services share a Victorian Fire Investigation Policy and Procedures agreement with Victoria Police, Victoria Forensic Science Centre, Office of Gas Safety, Energy Safe Victoria and WorkSafe Victoria regarding an integrated approach to fire investigation. The approach aims to identify trends and improve the targeting of fire prevention and community safety strategies.

2.11 Community education, awareness and engagement to prevent and respond to bushfire

Bushfire community education, awareness and engagement programs are a suite of specific interventions consisting of various programs that are intended to increase people's knowledge and understanding of bushfire risk and ultimately generate changes in behaviour that lead to a reduction of their exposure to bushfire risk. The extension of these programs in part aims to encourage disaster resilience within communities and to provide accessible, accurate and authoritative information. This empowers people to become more self-reliant and better prepared and is a key element of Victoria's shared responsibility approach.

Fire and land management agencies develop and deliver a variety of programs that take many forms reflecting the differences in individual and community needs, different risk environments, priorities and values. Current programs can be accessed online (CFA, DEECA or FRV). These programs within the Community Engagement and Education area are designed to prompt behaviour change within the Victorian community through knowledge transfer and information exchange.

The Department of Health and the Environment Protection Authority have important roles in supporting community members to manage impacts from bushfire smoke.

3. Planning

3.1 State, Regional and Municipal Fire Management Planning

The EM Act 2013 sets out a range of planning responsibilities:

State Fire Management Planning

At State level, the EMC is responsible for preparing the SEMP to provide for an integrated, coordinated and comprehensive approach to emergency management across Victoria. The SEMP is approved by the State Crisis and Resilience Council (SCRC).

This plan is a sub-plan to the SEMP and provides more detail regarding the specific arrangements for the management of bushfires in Victoria.

Updates to the SEMP Bushfire Sub-Plan are led jointly by the fire services agencies (DEECA, CFA and FRV) on behalf of the Bushfire Risk Mitigation Committee which has responsibility for oversight of updates to the SEMP Bushfire Sub-plan. The SEMP Bushfire Sub-plan is approved by SCRC.

Regional Fire Management Planning

At regional level, Regional Emergency Management Planning Committees (REMPCs) are responsible for regional emergency management planning, including the development of Regional Emergency Management Plans (REMPs).

REMPCs may adopt a fire management planning sub-committee or similar to provide technical advice and guidance relating to the management of fire to inform the development of a REMP and/or to prepare a fire management sub-plan to the REMP under Part 6A of the EM Act 2013.

CFA, DEECA and FRV are legislated members for all of Victoria's REMPCs. It is important that the fire services agencies contribute their subject matter expertise to regional level planning.

Municipal Fire Management Planning

At local level, Municipal Emergency Management Planning Committees (MEMPCs) are responsible for preparing Municipal Emergency Management Plans (MEMPs).

MEMPCs may adopt a fire management planning sub-committee or similar to provide technical advice and guidance relating to the management of fire to inform the development of a MEMP and/or to prepare a fire management sub-plan to the MEMP under Part 6A of the EM Act 2013.

CFA and FRV are legislated members to MEMPCs that are wholly or partly in the fire agencies footprints, and DEECA is able to be invited as a standing committee member at a MEMPCs discretion.

It is important that the fire services agencies contribute their subject matter expertise to local level planning.

The SEMP outlines the roles of the EMC, REMPCs and MEMPCs in further detail, including function and composition.

3.2 Strategic Bushfire Management Planning

Strategic bushfire management planning is undertaken to develop regional Bushfire Management Strategies. This process identifies where important values and assets are located across the landscape. It considers the current extent and quality of these values and where possible considers future trends including population, industry and environmental change. Strategic planning identifies objectives for the important values and assets and develops an approach to manage the risks posed to them. The resulting regional Bushfire Management Strategies describe landscape approaches to deliver bushfire risk reduction and improved ecological resilience outcomes.

Strategic bushfire management planning brings together land and fire managers, communities and stakeholders to develop a common understanding of bushfire risk and determine strategies and actions to reduce that risk.

Strategic bushfire management planning is jointly delivered by FFMVic, CFA, FRV, EMV, Department of Transport and Planning and councils in consultation with communities.

The key output of strategic planning are six Bushfire Management Strategies aligned to Victoria's Emergency Management Regions:

- Barwon South West
- Gippsland
- Grampians
- Hume
- Loddon Mallee
- Metropolitan (a joint strategy for Eastern, Northern and Western, and Southern Metropolitan regions)

The regional Bushfire Management Strategies focus on fuel management options which involve the modification of bushfire fuels such as leaves, bark, twigs and shrubs from the landscape. It can include activities such as planned burning and mechanical treatments including mowing, slashing, mulching and using herbicides. Each regional Bushfire Management Strategy includes a Fuel Management Strategy which outlines the arrangement of Fire Management Zones (FMZ) on public land, and a Bushfire Risk Engagement Area (BREA) strategy which covers both public and private land.

Strategic bushfire management planning for other bushfire management actions, including prevention of ignition and first attack suppression, has been piloted. Further improvement is being undertaken in this area and future Bushfire Management Strategies will include an expanded range of risk reduction strategies.

Strategic bushfire management planning enables agencies and communities to set clear objectives and devise strategies to achieve them. It results in a common understanding of risk, and approved strategies for reducing it. Regional Bushfire Management Strategies inform more detailed operational-level planning, including municipal fire prevention planning, CFA and FFMVic JFMP, and readiness and response planning.

The community is informed about regional Bushfire Management Strategies through an interactive online mapping platform and the publication of the strategies.

3.3 Forecasting risk

Fire services agencies use Fire Behaviour Analysts (FBANs) and bushfire risk analysts to combine meteorological and fire science inputs to develop fire behaviour predictions and assess where and when bushfire risk occurs. The predictions help fire management agencies to identify the potential likelihood and consequences of bushfires and help inform the development of strategic bushfire management, risk mitigation and public safety strategies.

In 2015, Victoria introduced a risk-based approach to fuel management on public land. Intelligence gathering and forecasting systems and techniques have allowed a greater quantification of bushfire risk. Victoria has adopted a system that allows simulation and assessment of many thousands of potential bushfires. This allows identification of the likelihood of bushfire occurring across the landscape and where the greatest potential impact will occur, such as loss of life or property.

A risk-based approach allows bushfire management activities, such as fuel management, to be targeted to reduce risk most effectively.

The drivers for bushfire risk differ across the State, from the types of forestsand other vegetation, the topography, and the location of communities. One important driver of bushfire risk is the fuel hazard levels across the landscape. The modelled fuel hazard data for Victoria is derived from analytical models which consider the fuel types (vegetation types), fire history and fuel accumulation equations.

The community is informed about bushfire risk through the publication of clear, meaningful information through multiple channels, such as Victoria's Bushfire Risk Management Report. EMV also communicates bushfire risk information through public media.

AFAC publish bushfire seasonal outlooks developed in consultation with the Bureau of Meteorology and fire and land management agencies. Seasonal outlooks are published in the last month of the preceding season for winter, spring, summer and autumn. Over the last several years, fire and land management agencies have worked closely with the Bureau of Meteorology in developing new capability and science to more accurately predict the physical drivers behind fire weather conditions. This new capability allows the prediction of fire weather and fuel state from multi-week to seasonal time scales at high spatial resolution.

3.4 Local Mutual Aid Plans (LMAPs)

LMAPs are developed by the fire agencies on an annual basis for each emergency management region in accordance with a Joint Business Rule. These plans detail agreed arrangements and procedures that support and enable a consistent, effective and integrated approach to bushfire readiness and response. LMAPs are required to meet the preparedness arrangements specified in JSOP02.03 – Incident Management Team Readiness Arrangements.

3.5 Public media campaigns

Fire services, often in partnership with Government, design and deliver media campaigns as a major component in the suite of activities and programs that support community safety in the context of bushfire. Media campaigns provide an outlet to get information to a large audience in a practical and timely manner. All forms of media are utilised that range from community and local information to State-wide campaigns, with dedicated efforts made to support and inform CALD communities through translation services and partnering with trusted community networks. Media includes television and radio, as well as press advertising, social media and feature articles in print.

A multi-agency State-wide bushfire media campaign is conducted annually in Victoria. The campaign is phased to reflect the fire season with the initial phase usually commencing in November. The campaign includes key safety messages and information encouraging residents to participate in other programs and activities and to get information from the agency websites and dedicated hotline, the VicEmergency Hotline. Local and regional press are heavily utilised to disseminate information. The next phase, which runs during the fire season, more specifically targets residents in high bushfire risk areas and promotes ways to access up to date information during bushfires.

Ultimately the outcomes sought from these activities include the increased awareness and acceptance of the bushfire risk, particularly amongst members of the public who are exposed to an elevated level of bushfire risk. This awareness is generally accepted as a necessary pre-cursor for residents living in fire-prone areas to find out more locally specific information, undertake preparedness activities and act to mitigate their risk.

4. Preparedness

There are a range of activities carried out by the emergency management sector, individuals and households, businesses and community groups to fulfil their shared responsibility in preparing for bushfires. The State provides a variety of supports to facilitate this.

All Victorians, no matter where they live, must understand their bushfire risk. They have a responsibility to learn about bushfire and to undertake measures to mitigate their own exposure to it. They must act to ensure their own safety.

Bushfire safety involves effective planning and preparation prior to a bushfire, making informed decisions during the event, and having access to a range of safety options, in particular places to shelter from the effects of the fire.

4.1 Community preparedness

The Bushfire Safety Policy Framework is reviewed annually to provide government, agencies and other stakeholders with direction on the development and implementation of bushfire-related public safety policies and programs. It reflects the findings and recommendations of the 2009 Victorian Bushfires Royal Commission and is informed by the latest bushfire research.

The Bushfire Safety Policy Framework establishes the following context for bushfire-related public safety policies and programs:

- The protection of human life is paramount.
- Risk management is fundamental to bushfire safety.
- Bushfire safety is a shared responsibility between the government and a range of stakeholders. However, individuals are ultimately responsible for making their own decisions about how to respond to the bushfire risk.
- The State has a duty in equal measure to prepare for, prevent, mitigate, respond to and support recovery from bushfire. The State cannot absolutely guarantee the safety of all its citizens from bushfire.
- All bushfires are different. Bushfires are complex and dynamic events. Safe responses will always depend on specific circumstances, so agencies and the community need to plan for a variety of situations.
- Bushfire safety depends on people having access to a range of safety options.
- All options other than being out of the fire area involve varying degrees of danger.

The State will provide timely advice to the community of forecast dangerous fire conditions and will warn communities wherever possible if a fire is likely to affect their location so that people are able to make informed decisions about how to respond.

Bushfire safety information, warnings and other safety interventions must recognise the diversity in the community and be accessible, address different needs and be relevant to local situations. The State will endeavour where possible to inform visitors to Victoria of the local bushfire risk and how best to ensure their safety.

The Bushfire Safety Policy Framework aims to create safe and resilient communities through setting directions for:

- increasing the level of public understanding and preparedness for bushfire
- enhancing the ability of those at risk from bushfires to make informed decisions about how to respond
- enabling safe response during bushfires by establishing a range of bushfire safety options.

4.2 Community resilience

The National Strategy for Disaster Resilience provides high-level guidance on disaster management to federal, state, territory and councils, businesses, community leaders and the not-for-profit sector.

Foremost in the Strategy is the principle of all of society taking responsibility for preparing for disasters. The application of this strategy within a bushfire context includes:

- individuals taking responsibility for their own safety and actively planning and preparing for bushfire
- councils and communities conducting local fire prevention and preparedness programs
- industry, including critical infrastructure providers, tourism and agriculture, recognising the potential risk of bushfire on their businesses and planning for continued service provision
- non-government and community organisations, to which the community may turn for support or advice, preparing for major bushfires
- government agencies through:
    - applying risk-based land management and planning
    - creating organisational partnerships to build community capacity and capability
    - providing bushfire education
    - supporting individuals and communities to prepare to respond to and recover from bushfires
    - ensuring an effective, well-coordinated response to bushfires
    - providing information to the community during bushfires
    - helping communities to recover and learn following bushfires.

The community, businesses and individuals can better withstand and recover from a bushfire through early financial recovery. Having insurance that adequately covers the different types of losses that can be experienced from a bushfire, is a key factor in supporting early financial recovery.

The National Recovery and Resilience Agency helps support local communities respond to large-scale natural disasters and undertake new initiatives to manage the impact of future events and the changing climate.

4.3 Neighbourhood Safer Places–Bushfire Place of Last Resort (NSP-BPLR)

CFA works with councils in fire prone areas to identify 'Neighbourhood Safer Places'; they are known as a 'Bushfire Place of Last Resort' (NSP-BPLR). These are places of last resort for people to move to and seek safety from a bushfire when all other bushfire plans have failed. NSP-BPLR's are locations that may provide some protection from direct flame and heat from a fire.

These places of last resort are at an existing location and do not generally have a purpose-built, fire-proof structure. In the main, NSP-BPLRs consist of a clearing that provides separation distance from the bushfire hazard (e.g. forest). They are often simply an open space at a local sports field, foreshore or park, or they may be located within a community building such as a hall or sports pavilion.

Councils play a key role is establishing NSP-BPLRs which are included in fire management planning activities and are assessed by the CFA against strict guidelines to ensure designated sites meet a range of criteria NSP-BPLRs do not guarantee the safety of community members during a bushfire.

4.4 Community Fire Refuges

A Community Fire Refuge is a last resort shelter option. It is a designated building that can be opened during a bushfire to provide the public with short-term shelter from the immediate life-threatening effects of a bushfire. They are purpose-built or modified buildings that have been formally endorsed by the EMC and can provide protection from radiant heat and embers.

Victoria currently has five Community Fire Refuges at Millgrove, East Warburton, Ferny Creek, Blackwood and Lavers Hill. Each is a dual-purpose building, co-located with either schools or emergency services facilities and placed in communities with no other real last resort options in the event of fire.

Community Fire Refuges are one of a number of contingency shelter options contained in Victoria's Bushfire Safety Policy Framework and should be considered in the context of all of the survival options available to a community in a high bushfire risk area. The Bushfire Safety Policy Framework acknowledges that there will be circumstances in which people may need to seek a last resort shelter option because their plans, such as leaving early or defending a well-prepared home, have failed.

4.5 Community information

The CFA website provides community information and advice tailored to user location, sourced through a postcode or locality search on the Local Information webpage.

The types of local bushfire information available includes:

- The relevant fire district.
- Total Fire Ban status and four-day Fire Danger Ratings predictions in the area.
- Location of the nearest brigade and any Neighbourhood Safer Place (NSP-BPLR; see section 0).
- Upcoming meetings and bushfire safety events.

Previously published Community Information Guides are no longer available.

4.6 Business preparedness

Every business and farm should have a written bushfire survival plan that is well-practised (see section 0). Business are also encouraged to use the Tourism Business Fire Ready Kit as the basis for a plan to help understand risks, prepare businesses and employees, plan how to inform customers and identify triggers to act.

4.7 Household and individual preparedness

Each household is responsible for preparing their own property for bushfire, for example, clearing vegetation around the house, in accordance with State planning controls for the clearing of native vegetation and council requirements (see section 0).

The CFA has a range of programs (including meetings, home visits, phone and online services) to assist householders and landowners to prepare their property for bushfire. Information on these programs can be found on the CFA website.

Every household should have a written bushfire survival plan that is well-practised. CFA has a wealth of resources and publications to support households and individuals develop bushfire plans.

A private bushfire shelter (commonly referred to as a bushfire bunker) is an option of last resort where people can take refuge during a bushfire while the fire front passes. Further information, including on permit requirements, is available on the Victorian Building Authority website.

4.8 Australian Fire Danger Rating System (AFDRS)

The AFDRS is the new Fire Danger Rating System, which commenced on 1 September 2022, and is being used across Australia. This means that whether people are at home or travelling, the same Fire Danger Rating System will apply.

The AFDRS calculates, forecasts and reports fire danger using up-to-date fuel state data, spatial and satellite data, weather data, science and technology. For an overview of the AFDRS see: AFAC | Australian Fire Danger Rating System Overview.

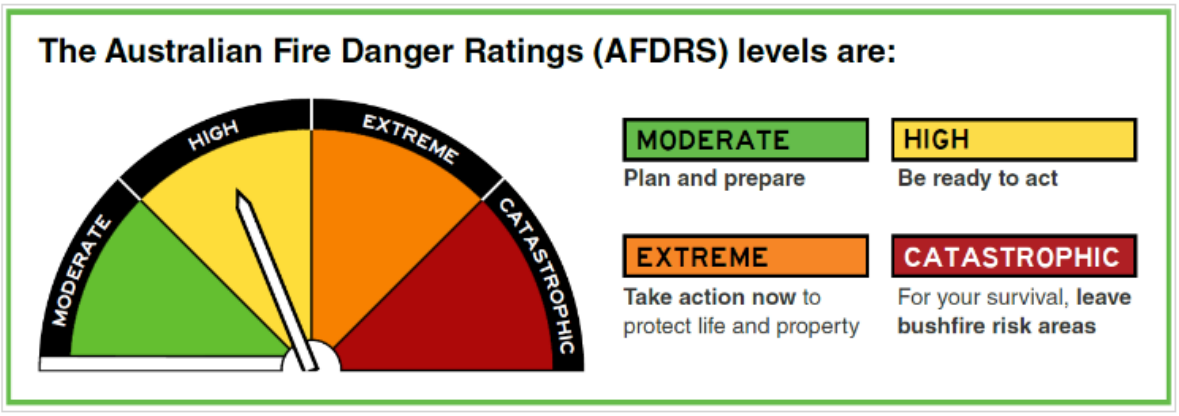
In the AFDRS, there are two main ways to look at fire danger: Fire Danger Ratings and the Fire Behaviour Index.

4.8.1 Fire Danger Ratings

The purpose of Fire Danger Ratings is to provide an efficient and easy way to communicate fire danger broadly to the community.

The new Fire Danger Rating System has four levels.

Figure 3: Sample Australian Fire Danger Rating Display



Daily forecast maximum fire danger ratings will be shown for the BoM Fire Weather District on roadside signs (where installed), emergency services websites, the BoM webpage and are broadcast where required through media such as television, radio, newspapers and the internet.

The Fire Danger Ratings feature:

- Broad categories representing levels of fire danger risk.
- Four fire danger rating levels.
- Simple fire danger rating names and intuitive colours.
- Distinct action orientated messages for each fire danger rating level.

An AFDRS Communications Kit for the community is published on the CFA website, available at: Australian Fire Danger Rating System Communications Kit | CFA (Country Fire Authority).

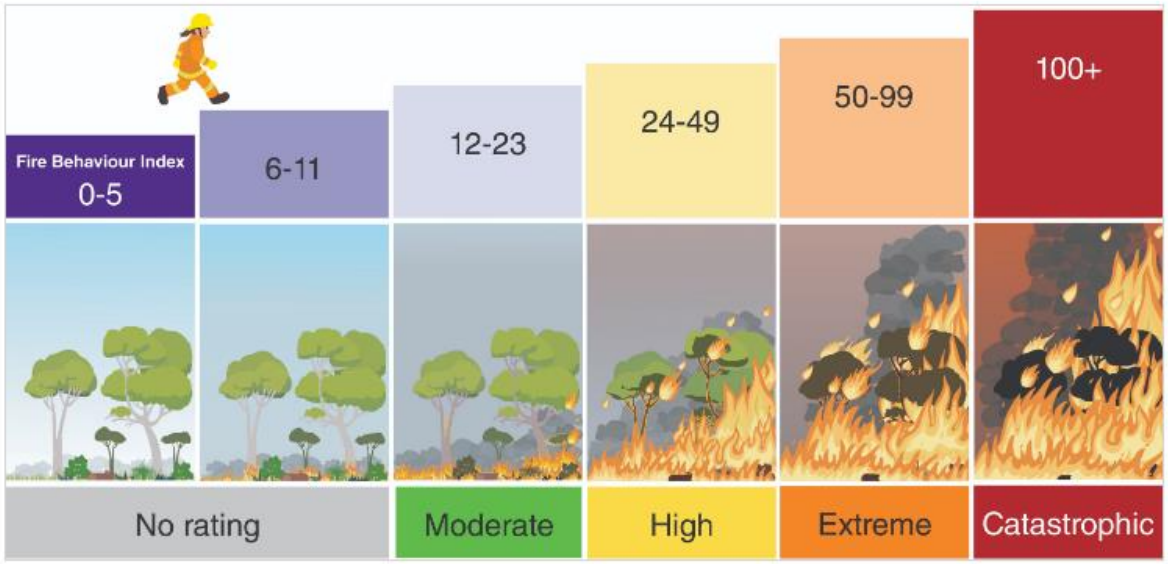
4.8.2 Fire Behaviour Index

The Fire Behaviour Index (FBI) is a simple numerical scale that can be used consistently across Australia, allowing users to make decisions that require finer detail than the four Fire Danger Rating categories allow. The FBI runs from 0 to 100 and beyond, with increasingly high values indicating increasing dangerous fire behaviour and therefore fire danger risk.

The FBI is designed to support fire management professionals and decision makers.

FBI forecasts are available on government agency intranet portals, through the registered users' section of the Bureau of Meteorology website and publically available during the high-risk weather season on the Bureau of Meteorology Victorian Fire Danger Ratings webpage.

Figure 4: Fire Behaviour Index scale



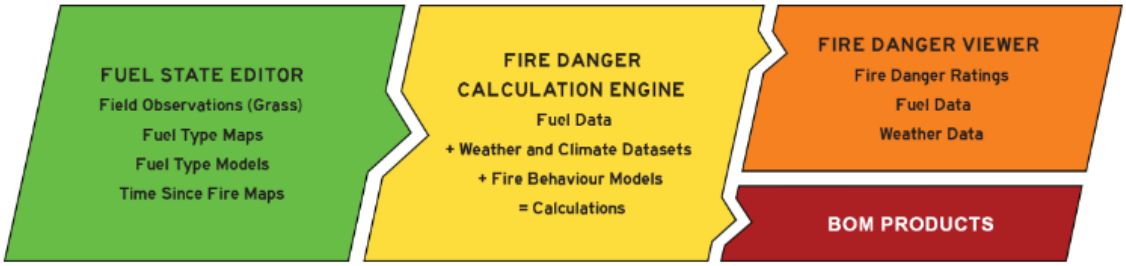
The Fire Behaviour Index features:

- A fine-grained scale of fire danger that runs from 0 to 100 and beyond.
- A stepped design to aid in operational decision making.
- Transition between steps based on changes in fire behaviour that have operational consequences.
- Management guidelines for each stepped category.

4.8.3 Components of the Operational Build

The Operational Build of the AFDRS is the underlying infrastructure that enables the latest science, technology and data to be used to produce accurate and timely fire danger information. There are four main components that work together to provide fire danger information.

Figure 5: Components of the Operational Build



Fuel State Editor (FSE)

The FSE is an interactive tool that allows fire and land management agencies to capture field information and update maps relating to the fuel component of fire danger; this data is then exported to the Fire Danger Calculation Engine.

Fire Danger Calculation Engine

This is the component that combines the various fire behaviour models with fuel, weather, and climate data sets to derive Fire Danger Ratings, a Fire Behaviour Index and other outputs on a seamless timescale. The calculation engine outputs data sets to various portals or feeds where users can see and/or interact with the information.

Fire Danger Viewer (FDV)

The FDV is an interactive and common visualisation tool for states and territories to view and interrogate fire hazard information; it includes the Fire Danger Ratings in map and table format as well as inputs such as weather and fuel data.

BOM Products

The Bureau of Meteorology (BOM) produces products as part of the AFDRS suite of services that provide nation-wide standard information about fire danger for a variety of uses.

Further information on the Operational Build including access to AFDRS eLearning courses is available at: AFAC | Australian Fire Danger Rating System Design.

4.9 Control Agency Preparedness

The EMC and the fire agencies prepare a suite of arrangements for responding to bushfire, including:

Ensuring a sufficient resource capacity for sustained response, including:

- A sufficient number and distribution of trained and experienced personnel
- Firefighting equipment, such as tankers and bulldozers, designed specifically for bushfire
- An aircraft fleet to support ground firefighters
- A network of airbases and communications infrastructure such as radio and fire towers
- Modern logistics systems to support firefighters.

Maintaining a science and technology capability to:

- Forecast fire risk and impact
- Provide firefighters with information about the fire.

Arrangements for the readiness of resources, including:

- Readiness plans
- Pre-positioning of resources.

Procedures for initiating response to fires, including:

- Bushfire detection systems
- Call-taking and dispatch arrangements.

In the lead up to each summer season, the EMC also requests assurance from fire services agencies and other key support agencies regarding their preparedness for the upcoming season.

4.10 Human resources

CFA has a strong volunteer workforce consisting of approximately 50,000 volunteers (with approximately 30,000 having an operational role in service delivery) supported by other staff who collectively deliver services across the emergency management spectrum for bushfire and other hazards.

DEECA works with FFMVic partner agencies to access approximately 1,800 staff with bushfire and emergency management skills during emergencies. DEECA employs seasonal project firefighters over the summer to complement permanent firefighting staff. FFMVic hosts significant specialised forest firefighting capability including staff trained and equipped to operate in remote settings, enabling rapid suppression and campaign fire response capabilities. FFMVic also provides approximately 1,300 incident management resources and support capabilities for bushfire at the State, regional and incident levels throughout Victoria.

FRV relies on a workforce of over 3,500 career firefighters supported by over 600 other staff.

Additionally:

- The fire services agencies have agreements with a wide range of agencies and interest groups regarding the delivery of specialist services during bushfires.
- The fire services agencies have pre-emptive agreements with comparative interstate and international agencies to provide reciprocal support during major fires and, in some cases, during preparedness arrangements.
- Arrangements are in place to source Commonwealth resources (such as the Australian Defence Force) if required.

4.11 Cross border arrangements

The Arrangement for Interstate Assistance (AIA) is the primary arrangement for mutual assistance in emergency management activities being conducted by Australian emergency response agencies. The AIA outlines processes for when significant resource deployments are requested by states to assist in response to large scale events, including bushfires.

Victoria has appointed a Cross Border Commissioner who works with border residents, businesses and community organisations to identify and advocate for change along Victoria's borders. This involves working with Victoria's departments, agencies and councils, helping them to Ask the Border Question on specific issues and across their portfolios of work.

The Cross Border Commissioner also works with interstate counterparts for cross-border advice and support, including the NSW Cross Border Commissioner and SA Cross Border Commissioner. A Memorandum of Understanding with NSW is now in place.

There are also a number of cross border agreements in place between Victorian emergency response agencies and their interstate counterparts, on both the New South Wales and South Australian border. The agreements cover first response, requests for immediate assistance and resource sharing with neighbouring jurisdictions.

The cross-border agreements can be accessed through the EM-COP Cross Border portal and through the relevant REMPC.

4.12 Capability

State capability and capacity is more effective and efficient as the emergency management sector has an all-agency, all-hazards approach where the agencies work as a unified, interoperable and agile team.

The fire agencies need to adequately respond to bushfire and other emergencies in order to effectively manage the impacts of such events on the community. The fire agencies undertake a range of planning and preparatory activities to ensure the State can respond effectively and efficiently to emergencies.

No single fire agency can manage a major bushfire alone. Agencies work together to provide a collective response force that builds on the strengths of individual agencies. Each agency strengthens the overall State capability and capacity to prevent or respond to bushfires by maintaining their own specialised skills and expertise related to the sphere of emergencies for which they have legislative responsibility.

The fire agencies actively work towards shared systems, common standards, consistent processes and comparable training of roles to better support efficient and effective operational activities.

The fire services train their personnel in a wide range of bushfire response, incident management and support roles. They conduct their training and exercises as joint–agency events wherever possible to ensure an effective integrated bushfire response.

4.12.1 Training, Equipment and Information Communication Technology (ICT)

Each response agency maintains its own vehicle and equipment fleet designed to meet its jurisdictional responsibilities.

Wherever possible, the fire services agencies collaborate during equipment design to improve interoperability for the future.

Where appropriate, the fire services agencies maintain a comprehensive register of contract equipment that can be hired during an emergency.

4.12.2 Shared ICT platforms

The EM sector uses several shared platforms to enable personnel to access information and systems across agencies. This includes shared storage, mapping systems, webmail, and incident management systems.

Emergency Management Common Operating Picture (EM-COP)

EM-COP is a web-based entry point for personnel across the emergency management sector to access information and systems using a standard web browser on any internet connected device.

EM-COP provides tabs for map-based collaboration, a desktop of links to current operational and other web-based information and a reference library. Access is via registered username and password and is open to all persons with a legitimate role in emergency management.

State Resource Request System

The multi-agency State Resource Request System (SRRS) is designed to make it easy for Resources Unit members at incident, regional and State tiers, to submit and action requests for additional resources during incidents and planned burning. The SRRS is a requesting system only and is not designed for resource tracking.

JSOP03.09 Resource Request Process sets out how to request resources for major emergencies.

4.12.3 Aviation

The fire agencies annually establish a fleet of contract and on-call aircraft, comprising both helicopters and fixed wing aircraft, to assist ground-based firefighters. Throughout periods of high fire risk, aircraft resources are strategically located at air bases around the State to support response activities.

Tactical and non-tactical aircraft undertake a wide range of operational and support activities and are crewed by appropriately trained and accredited personnel. Procedures detail readiness of aviation personnel and equipment in response to the risk environment. The procedures and contract arrangements provide for flexibility and scalability that provide for an adequate number and type of assets to be made available commensurate with the desired readiness level or meet the operational demand.

The State Air Desk operates as a function of the State Control Centre (SCC; see section 5.14.1) and coordinates the operational emergency aviation activity within the State for response activities under the control of the agencies. The Interagency Aviation Operating Procedures (IAOPs) are the procedures that govern the operation of aircraft for emergencies and land management activities. These procedures are produced to ensure the safety and effectiveness of agency aircraft operations by providing one source of agreed procedures which are available and relevant for all agency aircraft operations.

Once dispatched to an event, the aircraft are assigned to the incident and come under the control of the respective Incident Controller (IC).

The EMC is also able to request access to additional aircraft in the national fleet, coordinated through the National Aerial Firefighting Centre (NAFC). This national aircraft fleet complements aerial firefighting resources that are arranged directly by the State.

JSOP02.06 Aviation Resource Readiness (Bushfire) outlines the readiness levels of aviation resources.

4.12.4 Logistics

Each of the fire services agencies maintains an extensive store of logistical supplies so they can sustain a firefighting effort over a prolonged period.

The stores are:

- Fire Equipment Development Centre (DEECA) at Altona North
- State Logistics Centre (CFA) at Scoresby
- Sunshine Warehouse (FRV).

Arrangements are also in place with suppliers to obtain additional logistical supplies at short notice.

4.13 Intelligence and Predictive Services

4.13.1 Bureau of Meteorology

The Bureau of Meteorology (BoM) under agreement with the State's fire agencies produce a number of intelligence products that attempt to forecast fire weather and fire risk. BoM embeds a meteorologist in the SCC (see section 5.14.1). These products are informed by a number of data sources and published by BoM at regular intervals. Supplementary intelligence products that depict fire danger and bushfire risk are produced by the SCC and disseminated across the EM sector to inform decisions.

At a micro level, the fire agencies use fire behaviour analysts to combine meteorological and fire science inputs to develop fire behaviour predictions. The predictions help firefighters identify the potential impact and consequences of bushfires and help inform the development of bushfire control and risk mitigation strategies.

4.13.2 Intelligence products and Fire Behaviour Analysts (FBANs)

When a fire starts predictions of the potential impact of the fire are assessed at multiple levels including in the field by responding crews. To assist in more detailed analysis of fire spread potential, Fire Behaviour Analysts (FBANs) can be used to predict the areas of potential impact for up to several days in advance. The FBAN role relies on experience, predictive bushfire mathematical models and bushfire simulators across a range of fuel types to predict fire behaviour. The positioning of the FBAN role as part of the incident, regional or State management teams operating at the incident or State levels is relevant. The role is important for procuring reliable fire intelligence as well as providing early warnings (several days in advance) of largescale fire behaviour.

4.13.3 Remote sensing and other technology

The fire services agencies collaborate where possible on the development and use of technology to support firefighting. They primarily use technology to develop an operating picture of a bushfire that can be shared at the incident, region, State and community levels.

Technology such as remote sensing can also provide bushfire managers with a strategic overview of major bushfire.

5. Response

5.1 Concept of Emergency Operations

The State approach to the management of bushfire or any Class 1 emergency, consistent with the philosophy adopted Australia wide, is to ensure that:

- The protection and preservation of life and the relief of suffering is paramount
- Timely, relevant and tailored warnings and information are communicated to the community
- Agencies, which have personnel trained and equipped to provide a particular emergency response service, respond to the emergency
- All agencies are coordinated in their activities
- The provision of relief and recovery is integrated with response management at an early stage in the emergency
- There is ongoing assessment and management of the impact and consequences of the emergency
- The immediate needs of affected people and the impacts upon the community as a whole are managed
- State, regional and municipal plans are developed and executed, defining the operational needs for a multi-agency approach
- Control measures are in place at appropriate levels and an effective incident control structure is in place for all incidents to achieve protection of life and property
- Emergency Management Teams at State, region and incident levels are effectively led, managed and engaged.

These concepts apply to the response to an emergency, regardless of the size of the emergency, and regardless of how many agencies are involved in the response.

5.2 Tiers of Emergency Management

In line with the SEMP, Victoria has three operational tiers (incident, region and State) with the option of an 'area of operation' being declared to manage a complex emergency that may be geographically located over several municipalities or several regions.

Some bushfires have implications beyond the incident tier that require specific actions: they need more resources, have greater consequences and recovery needs or need messages sent to broader groups of people. In these cases, regional, State or area of operation arrangements may be enacted to support the incident.

Victoria's emergency management regions are declared by the Governor in Council under section 77A of the EM Act 2013. Region and State tier arrangements are constantly activated and scale up with resources and functional activities when a major emergency has occurred or is anticipated to occur through intelligence and commensurate with the desired levels of readiness.

The readiness and response structure for some emergencies may be enhanced by other arrangements or enhanced to cater for multiple emergencies of different types that may occur concurrently. The SEMP details how an Area of Operations may be established to meet the needs of a complex emergency; such an arrangement would operate in addition to the three tiers listed above.

5.3 Readiness

The State has a standing capacity and capability for responding to bushfires through standard agency arrangements. Agencies jointly develop local arrangements that document agreed arrangements for jointly preparing for and responding to bushfires.

Readiness arrangements are scalable and adaptable to ensure an efficient and effective response to bushfires of any size and complexity; they are often documented in a matrix style commensurate with the fire danger ratings system. Examples of adjusting agency readiness activities can include elevating arrangements when a significant emergency can be reasonably expected, such as a bushfire occurring on a day of elevated fire danger. Enhanced readiness arrangements can include activities such as pre-positioning incident management personnel, identifying surge resources, moving key assets such as aircraft to locations that better address the risk environment or increasing fire detection activities such as patrols, tower observers and fire detection flights.

5.3.1 Daily readiness arrangements

During the fire season, the fire agencies together with their partners within the State and Regional Control Teams (SCT and RCTs) jointly ensure response resources (personnel, equipment, facilities etc.) are set at appropriate level of readiness, commensurate with the risk and operational requirements. This can include pre-positioning fire-fighting resources and incident management teams on days of high fire risk.

At State and regional levels, a matrix for bushfire readiness summarise arrangements and guides decision making processes to ensure the appropriate capabilities are set to respond to outbreaks of fire quickly and effectively.

Fire agency readiness levels in respect to Incident Management Teams (IMT's), aviation and other specialist resources are recorded in operational systems that provide an overview or dashboard of readiness arrangements for bushfire.

The introduction of the AFDRS (section 4.8) provides a consistent basis for agencies and jurisdictions across Australia to plan and implement readiness arrangements for bushfire. This decision making is supported by other inputs including forecast fire danger indices (forest and grass), resource availability, existing fire activity and forecast lightning activity.

5.3.2 Resource readiness

In respect to some specialist resources, the fire agencies have an agreed set of arrangements for ensuring a minimum level of resources are ready to respond to bushfire emergencies in Victoria. JSOPs document these arrangements in respect to both IMTs and aviation.

During the fire season, aircraft are positioned at various locations around the State to support response activities. JSOP (JSOP02.06 – Aviation Resource Readiness (Bushfire)) for aviation readiness relates to the pre-positioning and readiness of aviation personnel and equipment to support bushfire response activities. These arrangements specify a minimum specification that can be adjusted and enhanced to suit the environment and other factors and the approvals required for this.

IMT and Incident Control Centre (ICC) arrangements in the fire season is specified in JSOP02.03 – Incident Management Team – Readiness Arrangements.

The approach to IMT readiness does not include the arrangements that control agencies may put in place to manage initial response and day to day activities, such as non-major fires, or normal fire response.

5.3.3 Operational planning

Operational planning and readiness arrangements informed by an environmental scan, resource availability and risk assessments are detailed and summarised in key operational planning documents.

The State Operations Plan is produced at the State tier and amongst other information provides a risk overview across Victoria for a defined period that can be used by fire agencies to inform readiness levels. The State Operations Plan also provides an operational overview, key State objectives and risks, and key strategic communication messages and activities.

Regional Control Teams (RCT) may produce a Regional 7 Day Action Plan relevant to each week across the fire season that considers similar information applied at the local level and considering the local risk environment. This Action Plan, approved by the Regional Controller (RC), describes and details the specific risks and activities of the fire agencies and support agencies in respect to bushfire readiness. It can be used to inform stakeholders of the necessary readiness arrangements in respect to bushfire risk for that planning period.

5.4 Initial Response

Following the ignition of a bushfire, the fuel, topography and weather in the area where the bushfire is burning will determine its size, direction and intensity. The fire agencies are responsible for the first response to bushfire according to their respective legislative and jurisdictional responsibilities, often supported by privately owned and managed resources in rural areas.

In light fuels such as grass, firefighters generally extinguish bushfires using water. Where water is scarce or where fuels are heavier, such as in a forest, firefighters generally contain bushfires within mineral earth firebreaks, created using either hand tools or heavy machinery.

Aircraft can assist in halting the intensity and spread of a fire for a temporary period, allowing ground firefighters to approach with increased safety.Predetermined dispatch of aircraft is a key measure to achieve this initial reduction in bushfire development.

5.4.1 Initial attack

Early detection of and response to bushfires is vital to restricting the spread and development of bushfires. The fire agencies detect bushfires using a range of methods including reports from the public, ‘spotters’ in fire towers, detection aircraft and observation by government employees during their daily activities. Enhanced levels of fire detection arrangements are activated commensurate with the forecast fire danger (see section 4.8).

The Emergency Services Telecommunications Authority (ESTA) provides the link between the Victorian community and the State’s emergency services agencies; they provide Victoria’s 24-hour emergency call taking and dispatch services and receive Triple Zero (000) calls from the public to report fires and dispatch CFA or FRV resources. Where a fire is detected on public land, ESTA dispatches the assigned CFA or FRV resources and contacts the FFMVic State Agency Commander who may dispatch additional FFMVic resources.

Where response is undertaken quickly to a bushfire, fires can generally be contained before they reach their maximum potential. An efficient first response will keep the area of impact of the bushfire as small as possible and will minimise the potential for bushfire to have broader consequences. On days of elevated fire danger, fire agencies adjust their readiness and response arrangements by planning for the initial response or first attack to fail and the fire to grow rapidly.

Once a bushfire has been contained, firefighters must ensure they consider and manage the risk of bushfire escaping containment. This will normally involve careful monitoring of the weather and on-going patrol of the area for several days to ensure there is no re-ignition of fuels previously thought extinguished.

5.4.2 Pre-determined dispatch of aircraft

At specific fire danger trigger levels, the State Aerial Firefighting fleet has a number of firebombing aircraft that provide pre-determined dispatch to have aircraft working over fires in the shortest possible time – rapid initial attack.

5.4.3 Operational Safety

In line with the State Emergency Management Priorities the safety and the protection of human life, including emergency services personnel and community members, takes priority above all other considerations in bushfire management.

All emergency services personnel are to avoid putting themselves at risk when working at an incident. By adopting the Safe Person Approach (SPA; see section 5.4.4) and using the Dynamic Risk Assessment (DRA; see section 5.4.5), emergency personnel may minimise the risk of injury to themselves and others.

5.4.4 Safe Person Approach (SPA)

Applying the SPA, agencies have responsibility to make sure systems of work are in place which allow work to be undertaken safely and, equally, personnel have a responsibility to ensure they work in accordance with agreed protocols. Each person has a responsibility for ensuring that their work practices do not result in an unacceptable level of risk to themselves or to others around them.

5.4.5 Dynamic Risk Assessment (DRA)

DRA is a simple continuous risk assessment process that allows operational personnel to rapidly and effectively assess risk, to decide on appropriate actions and controls and ensure responder safety in the rapidly changing operational environment.

When assessing risk, emergency services personnel safety is paramount.

DRA applies to all hazards associated with emergency response and operational activities and is carried out by all personnel from all agencies whenever there is an unexpected change to the plan or work environment. DRA is an intuitive thought process and is typically not recorded.

JSOP08.02 – Dynamic Risk Assessment outlines the practice of continuous DRA as a component of all operational activities, including emergency response and training, to ensure responder safety is maintained.

During established incidents, documented risk assessments are used to assess the options developed for fire containment and control.

5.4.6 Red Flag Warnings

Red Flag Warnings is a process used by fire agencies for communicating critical safety information to all incident personnel, to support decision making regarding strategy, tactics and deployment of resources.

JSOP03.11 – Red Flag Warnings details this process.

A Red Flag Warning is a message issued when there is a significant change to any critical information that may adversely affect the safety of personnel located at an emergency.

A Red Flag Warning should be issued when there is, or is predicted to be, a significant risk to safety due to changed circumstances, including but not limited to:

- weather conditions
- incident conditions and/or behaviour
- equipment availability
- communications arrangements
- access.

5.4.7 Field Information Update (FIU)

The FIU is a method for the distribution of changes to strategy, tactics, new intelligence, safety information to the field during a shift. It is typically new information that was not included or available at the time of the briefing. It may be delivered via radio broadcast or printed copy, and may include:

- weather
- incident behaviour
- organisation (sectorisation, control)
- communications.

5.4.8 LACES

In emergencies like bushfires, on ground responders rely on the use of safety zones and escape routes for safety as conditions change. In these situations, LACES provides a system to maintain safety. Alternate systems of work may be available to manage particular risks encountered during emergency work. If an alternative system is not available, or is unsuitable, LACES should be considered as part of the Dynamic Risk Assessment process (see section 5.4.5).

L	OOKOUTS
A	WARENESS
C	OMMUNICATIONS
E	SCAPE ROUTES
S	AFETY ZONES

5.4.9 Health Monitoring

Health Monitoring helps mitigate health risks to responders and identify those who may be at risk of adverse health outcomes as a result of their response to an emergency incident.

It includes taking vital signs, and the assessment of presenting symptoms and relevant medical history to determine if a person's health is impacted by their involvement in incident response.

All personnel attending and working at bushfires are strongly encouraged to undertake health monitoring to support their health and welfare, however health monitoring is not mandatory except in circumstances where the Standard for Managing Exposure to Significant Carbon Monoxide Emissions - Responder Health states that all personnel are required to undertake health monitoring.

JSOP08.05 – Health Monitoring – Emergency Personnel for further information and the process for requesting Health Monitoring.

5.4.10 Hazard Trees

Tree hazard is a major risk to firefighters and responders at all stages of operations, from active suppression, mopping up and patrol, through to recovery. All reasonable steps must be taken to protect firefighters and responders (including those providing relief and recovery functions) from tree hazards. Awareness and identification of trees which present a hazard need to form part of ongoing dynamic risk assessment performed by all personnel. During an attack on a going fire, personnel need to be particularly vigilant in identifying hazard trees and treating any unacceptable risk.

Following the passage of the fire, hazard trees within striking distance of access/control lines or transport corridors require assessment and treatment as soon as possible, and before the commencement of any mop-up/blacking out/patrol. Only appropriately qualified or endorsed personnel can assess, mark and treat hazardous trees on the fire ground (including staging/briefing/assembly points), where practicable. Hazardous trees often cause roads and access around bushfire affected areas to remain closed for extended periods until assessment and treatments can be completed to mitigate these hazardous tree risks.

JSOP08.03 – Tree Hazard – Fire has developed to mitigate the risk to emergency services personnel.

5.4.11 WATCHOUT

WATCHOUT is an acronym used to remind firefighters of potential dangers to their safety and to give advice on safe work practices. Understanding the meaning of the acronym will help perform a more comprehensive risk assessment.

W	EATHER dominates fire behaviour, so keep informed
A	CTIONS need to be based on current and expected fire behaviour
T	RY OUT at least two safe escape routes
C	OMMUNICATE with your supervisor, your crew and adjoining crews
H	AZARDS beware of variations in fuels and steep slopes
O	BSERVE changes in wind speed, direction, temperature, humidity and cloud
U	NDERSTAND your instructions, make sure that you are understood
T	HINK clearly, be alert and act decisively before your situation becomes critical

5.4.12 Occupational Health and Safety (OH&S) Incident Reporting

Emergency Personnel are required to report OH&S incidents through the line of control (to their Officer in Charge/manager/supervisor/etc) as soon as practical. All incidents are to be recorded in the relevant agency's hazard and incident reporting system or by using OH&S Incident Card system (salmon card). All relevant sections of the OHS incident report card are required to be completed and forwarded to the Incident Controller or Logistics Officer. The Incident Safety Officer may also support this process (where appointed).

Minor OH&S incidents are to be resolved at the local level unless otherwise advised by the IC. When a serious injury has occurred to a member, contractor or member of the public as a result of agency response activities the IC is required to be notified as soon as reasonably practicable.

The IC is required to notify the relevant agency commander and this is required to be escalated through to the State Agency Commander (SAC). The notification will also occur simultaneously through both the line of control through to the IC, Regional Controller (RC) and the State Response Controller (SRC), and to the employer(s) of personnel involved in the incident.

When a serious injury, that is a WorkSafe notifiable incident, has occurred to emergency personnel, contractor or member of the public as a result of agency response activity, WorkSafe must be notified verbally within 24 hours by the home agency. Written confirmation on the approved form is required within 48 hours.

Where an individual agency has notified WorkSafe of a serious incident the agency is required to notify the IC as soon as practicable.

JSOP08.01 - Occupational Health &Safety (OHS) Hazard, Incident and Investigation reporting - Major Emergencies provides further information.

5.5 Control

5.5.1 Control Agency and Support Agencies

Control is the overall direction of response activities in an emergency, operating horizontally across agencies, including the tasking of other agencies as per the SEMP. Authority for control is established in legislation or in an emergency response plan and carries with it the responsibility for tasking other agencies in accordance with the needs of the situation. Only one agency can be the nominated control agency for any emergency. Similarly, only one Incident Controller (IC) can be appointed at any one time for a bushfire event.

Support agencies provide services, personnel, or material to support or assist control agencies or affected community members. Support agencies integrate into the broader operational structure by being included into the emergency management team operating at the respective tier(s).

Agency Chief Officers/Chief Fire Officers and the FRV Commissioner retain their existing statutory powers, authorities and responsibilities at all times, and are required to exercise these powers, authorities and responsibilities in accordance with the SEMP.

5.5.2 Determining the Control Agency

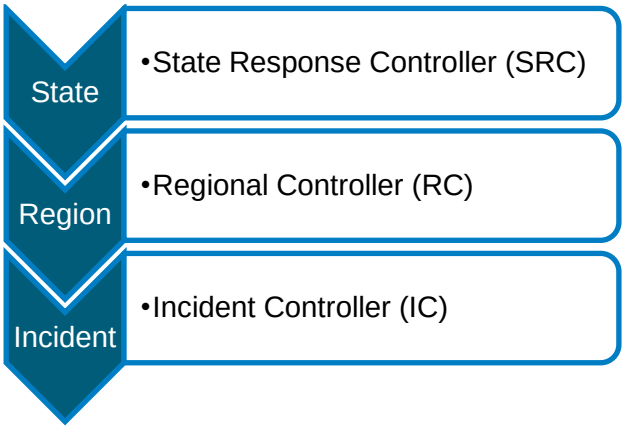
The fire services agencies are responsible for the first response to bushfires according to their respective legislative and jurisdictional responsibilities. As with each emergency and commensurate with the SEMP, each bushfire will have a single control agency and subsequently a single Incident Controller (IC). This is regardless of the number of agencies responding.

The control agency is determined by a number of factors, including initial location of the fire, predicted area of impact, and agreement between the fire agencies.

During initial response, the IC communicates to their agency through their agency command and control arrangements.

The ‘line-of-control’ refers to the line of supervision for those appointed to perform the control function in the context of a major emergency. For bushfire emergencies, the line of control is shown in Figure 6 Line of control in bushfire emergencies.

Figure 6: Line of control in bushfire emergencies



Additionally, and in order to meet the objectives of emergency management in Victoria, those performing the control functions need to ensure:

- the consequences of the emergency are managed, and
- there is communication that meets the information needs of communities, Government, agencies and business.

State Control Team (SCT)

In a Class 1 emergency, the EMC must appoint a State Response Controller (SRC) to be responsible for the control of response activities, in accordance with the SEMP. The SRC:

- is responsible for the control of response activities
- may exercise all the statutory powers and authorities of the chief
- officers of the relevant Control Agencies
- may appoint and deploy controllers in accordance with the SEMP
- is responsible for issuing warnings providing information to the community.

The SRC will establish a State Control Team (SCT) to provide advice on a strategic approach to the risk environment across the State.

The SCT for a bushfire usually comprises of:

- State Response Controller (SRC)
- State Agency Commanders (SACs; Class 1 agencies)
- Senior Police Liaison Officer (SPLO)
- State Recovery Coordinator (SReC)
- An Executive Officer will provide support to the team.

The SRC may request other functions to attend. Examples include the State Health Commander (Ambulance Victoria) or functional leads from the SCC such as Aviation, Public Information or the State Consequence Manager.

Regional Control Team (RCT)

The Regional Controller (RC) will establish an RCT to provide advice on a strategic approach to bushfire readiness and response and support the functions of the Region.

The RCT for a bushfire usually comprises of:

- Regional Controller (RC)
- Regional Agency Commanders (RACs; Class 1 agencies)
- Regional Emergency Response Coordinator (RERC)
- Regional Recovery Coordinator (RRC)
- An Executive Officer will provide support to the team.

The RC may request other functions to attend. Examples can include the Regional Health Commander (Ambulance Victoria), Commanders representing other agencies / departments or functional roles such as a Regional Resources or Intelligence Officer.

5.5.3 Transfer of Control

Establishing effective control arrangements in the early stages of the incident is critical. Control for a bushfire emergency will always be exercised at the incident tier. Once control is established, the Incident Controller (IC) can transfer control to another IC, provided this is done formally, a record is kept, and all involved agencies are informed of the transfer. The IC may be from another agency.

There are circumstances where an IC based in an Incident Control Centre (ICC) and supported by an Incident Management Team (IMT) with specialist skills and equipment should manage an incident, rather than a field-based IC. These circumstances include where the incident is a major emergency or has the potential to become a major emergency, or where the bushfire becomes

complex and there is a need to delegate the operation of a number of functions under the AIIMS system. In these transfer of control circumstances, the previous IC will generally connect the ICC with field operations by assuming a role in the Operations Section, such as a Division Commander. They can then focus on tasking field operations and providing information back to the IMT.

There are circumstances where the initial IC should transfer control of a bushfire emergency to an IC within a preformed IMT almost immediately. These include where:

- the incident shows clear potential to become a major emergency,
- initial attack has failed, fire weather is extreme or control is unlikely in the short term,
- the IC requires immediate specialist support,
- a number of similar incidents are expected within the area (such as multiple roof damage following a storm), best managed as the one incident.

Control may be transferred from one ICC to another more suitable ICC if it is better able to support the incident. This transfer may be required for various reasons, including personnel capability, functionality of the facility, escalation of the incident, or movement of the incident through the landscape. In some cases, a staged transfer of responsibilities to an IMT in an ICC may be the most appropriate course of action.

JSOP03.15 – Transfer of Control and IMT Relocation for Class 1 Emergencies, outlines the process for the transfer of control from the field-based IC to an ICC based IC.

5.6 Coordination

The EMC is responsible for the coordination of agencies at the State tier with roles and responsibilities for response to bushfire.

Victoria Police members are appointed as Regional Emergency Response Coordinators (RERCs) and Municipal Emergency Response Coordinators (MERCs) and undertake the coordination of resources at the regional and municipal / incident tiers of emergency management respectively.

Under the EM Act 2013, the Chief Commissioner of Police must appoint a Senior Police Liaison Officer (SPLO), who advises the EMC about regional and municipal emergency response coordination and other police matters and is the liaison channel with the RERC and MERC.

5.7 Command

Command is the direction of activities internally within an agency to use its people, resources, governance, systems and processes to discharge its responsibilities in line with relevant legislation. Each agency appoints agency commanders to supervise their personnel and ensure they are working safely.

The ‘chain-of-command’ refers to an agency’s organisational hierarchy that defines the accountability of people or positions and identifies the link between individuals and their supervisor. An agency might formally agree for a person from another agency to supervise their personnel for a fixed period during emergencies. However, this does not replace the home agency’s responsibility to their personnel.

5.8 Incident Management

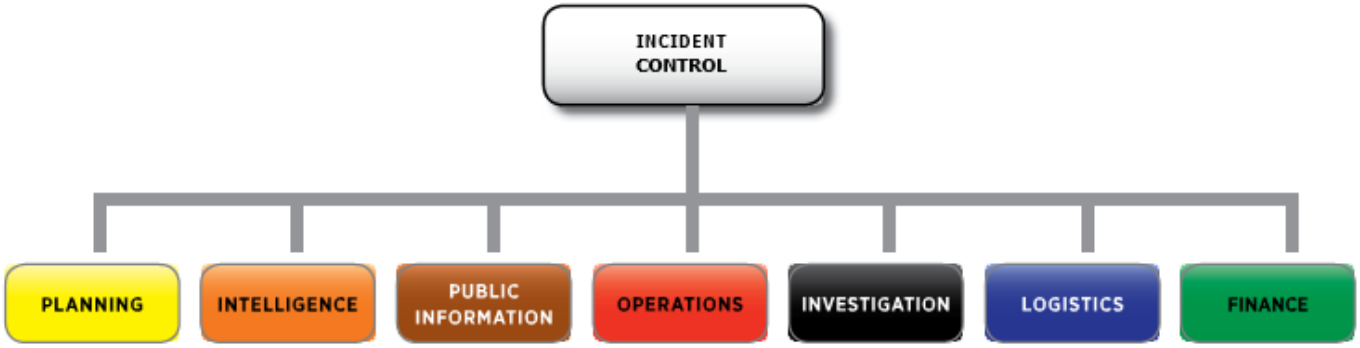
Bushfire will be controlled at the incident level through the application of the Australasian Inter-service Incident Management System (AIIMS). AIIMS uses the following principles to manage incidents:

- Flexibility
- Management by objectives
- Functional management
- Span of control, and
- Unity of command

AIIMS also applies the principles of risk management described in AS/NZS ISO 31000: 2009 Risk Management – Principles and Guidelines.

The IC is appointed for every incident and is responsible and accountable for all the functions of incident management. AIIMS identifies eight functional areas, as depicted in Figure 7 AIIMS Incident Management Functions and described in Table 3 Function Descriptions.

Figure 7: AIIMS Incident Management Functions



Source: AIIMS 2017

Table 3: Function Descriptions

Function	Description
Control	The management of all the activities necessary for the resolution of an incident.
Planning	The development of objectives, strategies and plans for the resolution of an incident based on the outcomes of collection and analysis of information.
Intelligence	The task of collecting and analysing information or data, which are recorded and disseminated as intelligence to support decision making and planning.
Public Information	Provision of warnings, information and advice to the public, and liaison with the media and affected communities.
Operations	The tasking and application of resources to achieve resolution of the incident.
Investigation	The task of conducting investigations to determine the cause of an incident and/or to determine factors that contributed to the impact of the incident or specific events.
Logistics	The acquisition and provision of human and physical resources, facilities, services and materials to support achievement of incident objectives.
Finance	The task of managing: • accounts for purchases of supplies and hire of equipment; • insurance and compensation for personnel, property and vehicles; and • the collection of cost data and provision of cost-effective analyses and providing cost estimates for the incident.

5.9 Incident levels

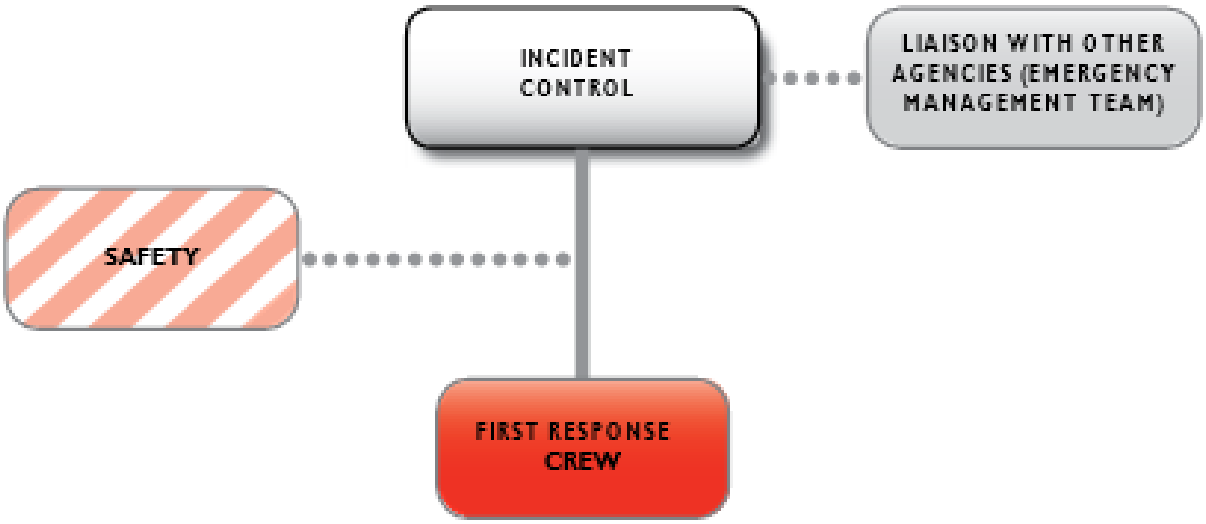
The classification of bushfire is based on the size, scale and risks of the incident and the resources needed to manage it. This classification allows the agencies to communicate the complexity of the incident and scale the response to suit the incident. Escalation of incident levels usually generates greater oversight by the region and state levels, which may include guidance on the classification.

5.9.1 Level 1 incident

Level 1 bushfire incidents are characterised by being resolved through the use of local or initial response resources only. Control is limited to the immediate area and the Incident Controller (IC) can usually perform all the necessary functions.

The IC may delegate some functions to personnel on scene (e.g. operations managed by crew leader) or remote to the incident (e.g. warnings issued by duty officer in district headquarters).

Figure 8: Example of Level 1 incident management structure



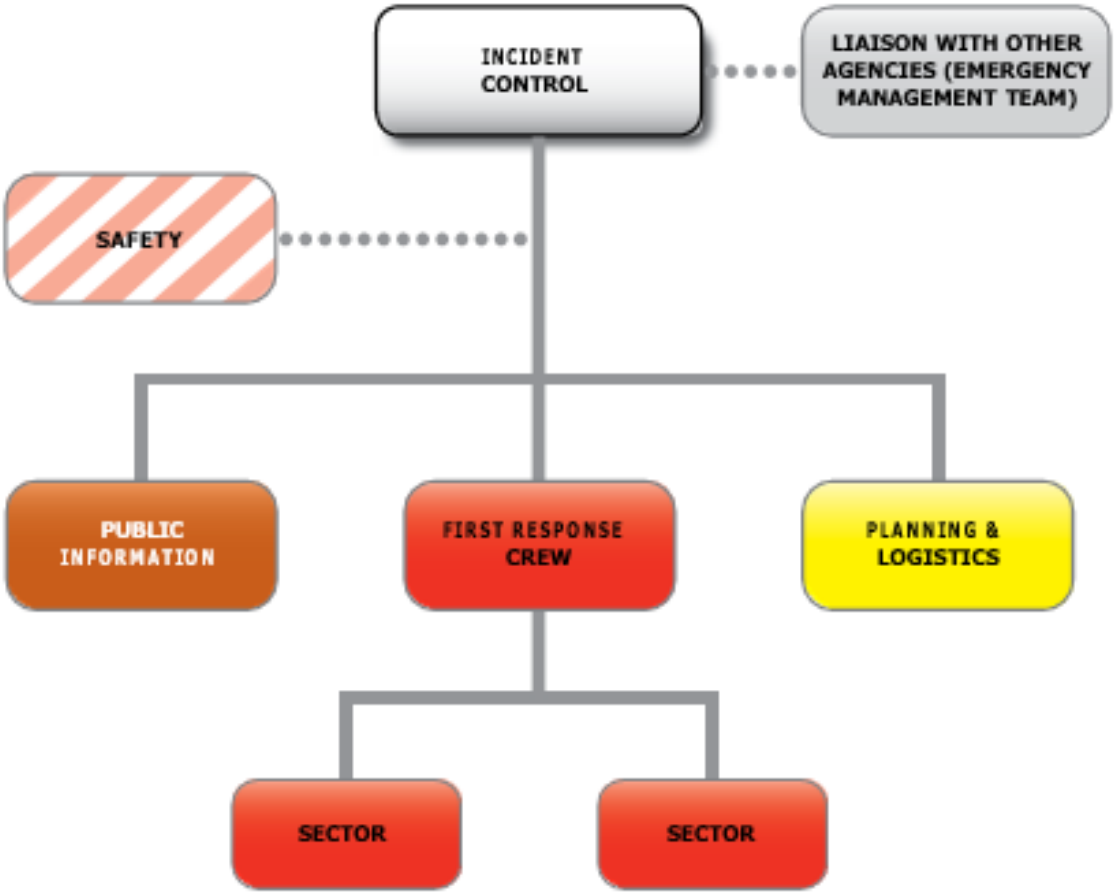
Source: AIIMS 2017

5.9.2 Level 2 incident

Level 2 bushfire incidents may be more complex either in size, resources or risk. They are characterised by the need for:

- deployment of resources beyond initial response; or
- the operations being divided into geographic or functional sectors; or
- the establishment of incident management functional roles due to complexity levels; or
- a combination of the above.

Figure 9: Example of Level 2 incident management structure



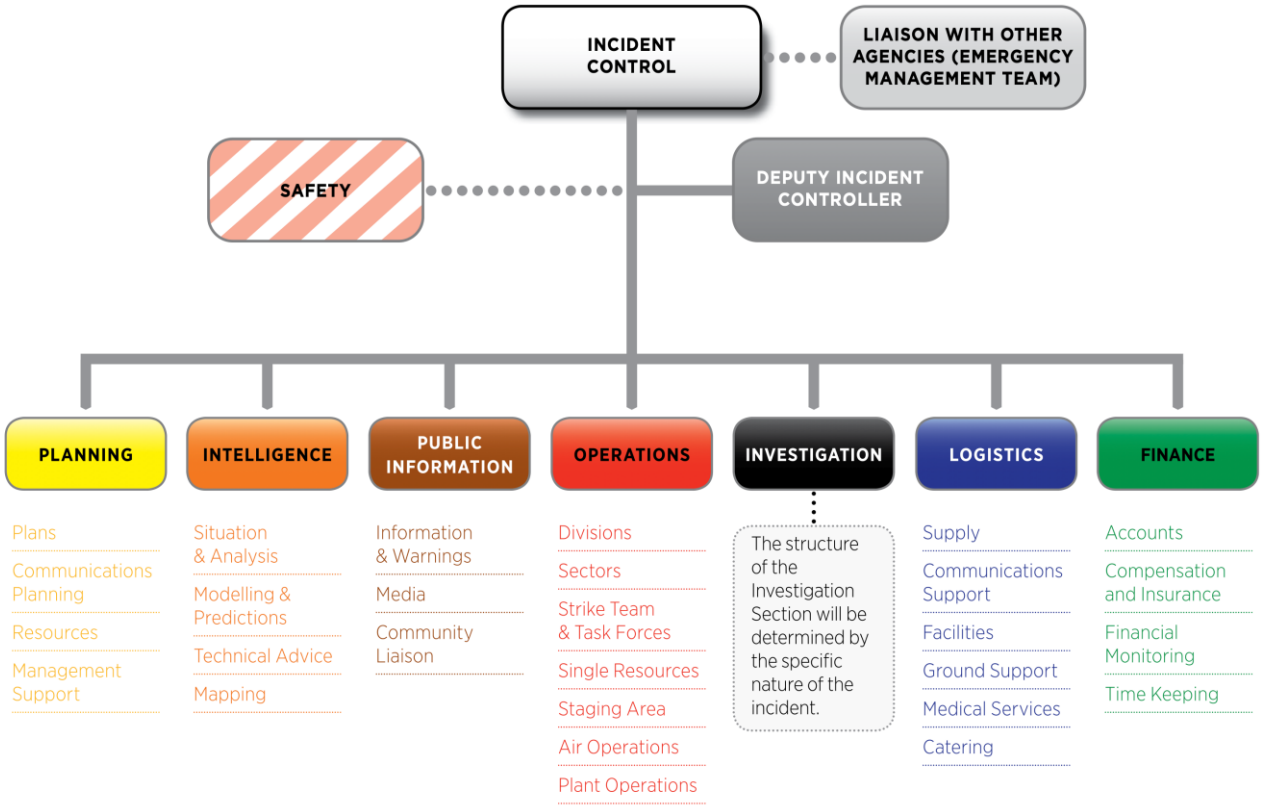
Source: AIIMS 2017

5.9.3 Level 3 incident

Level 3 bushfire incidents are characterised by degrees of complexity that may require a more substantial organisational structure to manage the emergency. These emergencies will usually involve delegation of all incident management functions and are considered major bushfire events.

An example of an incident management structure for a Level 3 incident is shown in Figure 10.

Figure 10: Example of Level 3 incident management structure



Source: AIIMS 2017

5.10 Incident Controller (IC)

The appointment, function, location and role of the IC is outlined in the SEMP.

As per JSOP03.08 – Appointment of Class 1 Regional Controllers and Incident Controllers, the control agency is responsible for the appointment of an IC for Level 1 and 2 incidents. If the incident is or is likely to become a major emergency the RC is required to ensure that an IC is appointed from the EMC's endorsed list of Level 3 ICs.

5.11 Incident Management Team (IMT)

Where required, the IC is supported by an IMT, Incident Emergency Management Team (IEMT) and any additional specialist resources. The appointment, function and composition of the IMT and IEMT are outlined in the SEMP.

There is only one IMT for each incident, at any point in time, regardless of the number of agencies in attendance. IMTs will be established and operate in accordance with the AIIMS incident management system.

5.11.1 Deputy Incident Controller (Deputy IC)

To support the management of the incident, the Incident Controller may appoint a Deputy Incident Controller(s) to have oversight of specific tasks or to cover times that the Incident Controller is absent from the ICC. The appointment of Deputy ICs ensures that the overall control of the incident can be maintained by a single person, the IC.

If appointed and required to do so by the IC, the Deputy IC will act as the IC in their absence; this is often the case for a night shift of a prolonged incident. Appointment, delegation and transition of responsibilities should be documented and communicated to the IMT and IEMT.

To ensure continuity, Deputy ICs may not alter the incident objectives in the Incident Action Plan. However, they may amend the incident strategies within the parameters provided by the IC.

5.12 Supplementary resources

The IC is responsible for identifying the resources needed to efficiently and effectively respond to the incident, sourcing these and establishing a safe area for them to stage prior to deployment to the incident. Local readiness, mutual aid arrangements, memorandums of understanding or local cross border arrangements between the control and supporting agencies should be used where established.

The SEMP's Requesting Additional Resources section provides further information.

5.12.1 Resource Requesting

The Incident Controller (IC) is responsible for providing direction, across all agencies, to the resources responding to the bushfire and for requesting and releasing resources. They are also responsible for identifying and sourcing the appropriate number and type of resources required to execute their Incident Action Plan. Local readiness, mutual aid arrangements, memorandums of understanding or local cross border arrangements between the control and supporting agencies are used to inform local resource allocation and deployment to an incident.

Once the IC has determined that all resources (from control and support agencies) within the district have been exhausted, resources may be requested by the State Resource Request System (SRRS) and escalated to the regional tier. The Regional Controller (RC) will seek out appropriate resources from within the region to fill regional or incident requests. When an RC cannot obtain resources within their own region, the RC will escalate the request to the State tier. JSOP03.09 Resource Request Process details this process.

Where the IC has requested further resources beyond the initial response, they should establish a staging area to assemble these as they arrive. A staging area assists the IC to control access and egress to the incident site, to brief resources prior to deployment, to assemble resources into work units (for example strike teams and task forces), to manage welfare needs and to ensure all the resources are working and communicating within the command and control structure.

5.12.2 Interstate and International Resources

Victoria is a partner in the Commissioners and Chief Officers Strategic Committee (CCOSC). When jurisdictions are likely to require additional resources outside of their jurisdiction, it directs the activation of the Australian Fire Authorities Council – National Resource Sharing Centre (AFAC NRSC). AFAC NRSC meetings are held as required and more regularly during times of national crisis. There are emergency management hubs across Australia which hold an annual expression of interest process for personnel to register their interest in being deployed interstate and international.

Victoria is identified as a partner under the Australasian Arrangement for Interstate Assistance - Fire and Emergency Services (AIA; see section 4.11). This is a mutual aid agreement between

Australasian fire services, emergency services and land management agencies. It is intended for use within Australia and between Australia and New Zealand.

Victoria has formal written agreements with Canada and USA which formally allow for the exchange of Wildland Fire Management Resources between these countries.

The induction of interstate or international personnel is critical to safety and functionality.

5.13 Emergency Management Team(s) (EMT)

The SEMP outlines the appointment, function and composition of Emergency Management Teams (EMTs) at each activated tier of emergency response management, for the function of response (including relief) and recovery coordination:

- State Emergency Management Team (SEMT)
- Regional Emergency Management Team (REMT)
- Incident Emergency Management Team (IEMT).

Not all agencies have the capability to provide a representative for EMT at each tier. For example, a person may represent their agency at both the REMT and IEMT.

5.13.1 Incident Emergency Management Team (IEMT)

The SEMP outlines the composition of an IEMT. Membership may also include:

- Support agency commanders (or their representatives)
- Health Commander (functional commander of supporting health agencies)
- Municipal Emergency Management Coordinators (MERCs) representing councils
- Utility providers / operators, and
- Other specialist persons as required.

Support agency commanders then implement the allocated strategy through their respective command structures, and report back to the Incident Controller (IC) as to the success or otherwise of the strategy.

The effective operation of an IEMT relies heavily upon communication between agencies. The importance of an effective IEMT to the successful management of an emergency cannot be overstated.

Although the IEMT facilitates a collaborative decision-making process, with the primary intent of unity and purpose of effort, the IC leads the team and retains control of the emergency.

5.14 Control facilities

A range of static or mobile facilities support the line of control by providing the appropriate physical and technical infrastructure. These may be activated in anticipation of an incident to monitor impacts in the potentially affected area.

5.14.1 State Control Centre (SCC)

The SCC is the State's primary control centre for management of emergencies and acts as a hub for a state-wide network of regional and incident control centres. The facility supports the State Response Controller (SRC), State Control Team (SCT), State Emergency Management Team (SEMT) and other key stakeholders.

5.14.2 Regional Control Centres (RCCs)

An RCC is the predetermined location where the Regional Controller (RC) operates from for a particular region. RCC functionality is maintained at all times to ensure immediate capability for all control agencies for all threats and hazards.

The RC, Regional Control Team (RCT) and other personnel that operate from the RCC form part of the line of control and chain of command structures, with direct responsibility for all ICCs and operations identified within their area of responsibility.

5.14.3 Incident Control Centres (ICCs)

An ICC is the location where the IC and the IMT manages response activities. An ICC may be established for any size incident to cater for the needs of the incident management structure. Each ICC should, where possible, be at a pre-defined location that allows an appropriate level of communication with personnel at incidents within its 'footprint'.

A Level 3 ICC is a facility used to accommodate an IMT during preparation for, or response to a large or complex (Level 3) incident. A Level 3 ICC will have all the facilities and services required to support the operation of a multi-agency IMT, including those facilities required by support agencies.

5.14.4 Control Point

A Control Point is a field location where the role of IC may operate from, often used during the initial stages of an incident and may be supported by purpose-built vehicles.

Regardless of the number of agencies at an incident, it is important that only one facility is identified as the place from which the incident is controlled. For larger scale or more complex incidents, control should transfer to an ICC.

Once control has transferred to the ICC, the original control point may be re- established as an Operations Point, Division Command Point, Sector Command Point or disbanded as appropriate.

5.14.5 Operations Point

An Operations Point is the location from which the overall field operations are commanded by the Operations Officer, once this role has been delegated. An incident will have only one Operations Point.

In large scale incidents, the Operations Officer will be located at the ICC in most circumstances. Where an ICC is established at some distance from the incident, the Operations Officer (in consultation with the IC may appoint a Deputy Operations Officer to attend the Operations Point.

5.14.6 Division and Sector Command Point

A Division Command Point (DCP) is a location where the person in the role of Division Commander operates. A Sector Command Point is the location from which the role of Sector Commander operates, if required.

A Division or Sector Command Point could be a mobile point close to the incident, a Field Command Vehicle (FCV) or a building such as a Local Command Facility (LCF).

5.15 Operational communications

Fire agencies need to ensure that Joint Default Communications Plans are prepared to cover their respective districts before each fire season. Local Mutual Aid Plans (see section 3.4) will detail these default communications plans that are to be implemented at each multi-agency incident, until the IC replaces it with a specific Incident Communications Plan; which may only occur to support complex bushfire operations.

JSOP02.02 – Incident Communications Planning sets out the process for the development, implementation and review of default and Incident Communications Plans.

5.16 Bushfire naming convention

Agencies are required to use a standard bushfire naming process to ensure that incidents are consistently named, quickly located and easily referenced. The incident name should be recognisable at all emergency management tiers and determined at a level as close to the incident as possible. The incident name is also a critical element of information provided to the community. The application of naming conventions ensures there is consistency in warning and advice messages.

The control agency is responsible for determining the incident name. For multi-agency incidents, agencies will ensure that their respective reporting systems show the correct incident name. Standard processes should be followed if incident name needs to be amended due to the escalation, merging, or other changes to the situation.

JSOP03.02 – Incident Naming – Major Emergencies provides further details surrounding the naming convention and requirements.

5.17 Traffic management

Traffic Management Points (TMPs) are established at the direction of the IC, in consultation with Victoria Police and Department of Transport and Planning, to regulate the flow of traffic into an area where bushfire has occurred, is occurring or has the potential to occur. Smoke, fire, hazardous trees, emergency services vehicles, wildlife and electrical assets are all hazards that can be reasonably expected on and around roadways in bushfire affected areas. For these reasons travelling on roads during or immediately after a bushfire can be particularly hazardous. Emergencies are not static and therefore the conditions of TMP may change over the course of and incident, and at any time.

JSOP03.10 – Traffic Management at emergencies outlines the process for the activation and ongoing monitoring and deactivation of traffic management points, including the appointment of a Traffic Management Manager, development of a Traffic Management Plan and associated forms.

Traffic Management Guidelines describe the agreed procedures for the operation of a TMP as part of an overall Traffic Management Plan, to assist in the control and management of pedestrian and vehicular road travel near an incident.

To facilitate entry into an area, a TMP will be assigned an access level. The circumstances and groups of people who may be permitted access at each access level will be determined by the Incident Controller (IC).

5.18 Evacuation

AIDR's Evacuation Planning Handbook defines evacuation as a risk management strategy that may be used to reduce loss of life or lessen the effects of an emergency on a community, prior to the onset of, or during, an emergency. It involves the planned movement of people threatened by a hazard to a safer location and, typically, their eventual safe and timely return. For an evacuation to be effective, it should be appropriately planned and implemented. Evacuations are not compulsory in Victoria and requirements for undertaking evacuations are detailed in JSOP03.12 – Evacuation for Major Emergencies.

The Incident Controller (IC) is responsible for making a decision as to whether evacuation is a safe option for communities and individuals. In making this decision the IC may seek advice from other agencies or communities.

Depending on the hazard and its impact and consequences on the community, the evacuation process, including withdrawal and return, may take days, weeks or months to complete. Some evacuations may be carried out very quickly and over very short distances. For example, people may be warned to move to higher ground pending a potential tsunami impact, to avoid a flash flood, or to move two streets away from a peri-urban interface to avoid a grass fire. In other circumstances, people evacuated from an area may be relocated many kilometres from their home and be unable to return for a considerable period of time.

Evacuation is a scalable activity in that it may be applied to individuals, a house, a street, a large facility (i.e. school or hospital), a suburb, a town or a large area of the State. Where an area is identified (by means of local knowledge, prior history of a higher risk of evacuation, etc.) as requiring a specific detailed evacuation plan, consideration should be made to include this plan as part of the respective Municipal Emergency Management Plan (MEMP).

5.19 Smoke

Exposure to smoke or emissions from fires, especially when prolonged, should be addressed to protect both workforce and community from potential health impacts, including:

- Health advice to protect both the responders (response, relief and recovery personnel) and the community
- For responders, deployment of monitoring equipment to the incident site to monitor composition and exposure
- For community exposure, potentially deploy monitoring where there are smoke impacts, in line with JSOP03.18 – Incident Air Monitoring and Advice for Community Health.

The following documents contain more information on smoke and the process for initiating monitoring and the provision of public information and warnings, including public health advice:

- State Smoke Framework[1]
- Standard for Smoke, Air Quality and Community Health – Significant fires with fine particles (PM2.5) as the primary smoke component of health concern
- Standard for Managing Exposure to Significant Carbon Monoxide Emissions - Community Health
- Standard for Managing Exposure to Significant Carbon Monoxide Emissions - Responder Health
- JSOP03.18 – Incident Air Quality Monitoring and Advice for Community Health
- JSOP03.19 – Managing Significant Community Exposures to Fine Particles and Carbon Monoxide in Smoke from Fires
- JSOP04.01 – Public Information and Warnings for Class 1 Emergencies

[1] The State Smoke Framework was developed following the Hazelwood coal mine fire with the last revision made in November 2016. Whilst the framework requires significant review, in-principle, the purpose of the Framework remains relevant and important.

5.20 Public Information

The provision of warnings and information in relation to fires in Victoria is detailed in the State priorities and also fire agencies have legislated responsibilities.

The EMC may also seek support from the Emergency Management Joint Public Information Committee (EMJPIC) to ensure the state-level messages from all agencies involved in the management of emergency response, recovery and the consequences of the emergency are included in the key messages to the public. Regional and Incident Joint Public Information Committees may be stood up if required or directed by the Regional Controller (RC) or Incident Controller (IC).

5.20.1 Provision of warnings to the community

Timely, relevant and tailored warnings and information must be issued to potentially affected communities. The Incident Controller (IC) is responsible for authorising all warnings and information provided to the public. To assist the rapid communication of warnings and information, the IC may authorise a Deputy IC or Public Information Officer (PIO) to authorise the release of warnings and information to the community. No additional authorisation is required once the IC or delegate has authorised the information or warning.

Where an extreme and imminent threat to life exists and it is not practicable to obtain authorisation from the IC in the circumstances, warnings may be initiated by any response agency personnel. The IC is required to be advised as soon as possible.

Victoria's emergency warnings arrangements include the use of Standard Emergency Warning Signal (SEWS), telephone alerts and message standards.

5.20.2 Warning levels and templates

Warning templates provide the Incident Controller (IC) and Public Information personnel the opportunity to include targeted information and specific actions for the communities affected. Warnings should be tailored to the community and reflect the impacts the bushfire is having or may have on the community. This includes information such as road closures, the direction the fire is moving in, where the community should go and information about what the community might see or experience; relief and support services, and health advice as required.

5.20.3 Levels of community warnings and information

The Australian Warning System has three warning levels, with a consistent set of icons for a number of emergencies, including bushfire.

Figure 11: Australian Warning System relevant to bushfire

Icon	Warning Level	Description
	Advice	An incident has started. There is no immediate danger. Stay up to date in case the situation changes.
	Watch and Act	There is a heightened level of threat. Conditions are changing and you need to start taking action now to protect you and your family.
	Emergency Warning	An Emergency Warning is the highest level of warning. You may be in danger and need to take action immediately. Any delay now puts your life at risk.

5.20.4 All Clear Messages

An All Clear message is to be issued to inform the community that they can return to normal activity that the risk from fire is over. An 'All Clear' message needs to be issued whenever a Warning (Watch and Act) or Emergency Warning has been issued. An All Clear – Safe to Return is required to be issued if Evacuation warnings have been issued.

5.20.5 Community Information

Community Information is used to provide the community with information if an incident is creating community interest but does not pose a threat to the community.

5.20.6 Publishing Warning and Information Areas (Polygon)

Warnings and information issued to the community will have the warning area (polygon) shown on the map published to the VicEmergency website and app. Incident Controllers (ICs) should ensure that the polygon associated with the warning or information captures the community who needs to act.

5.20.7 Community meetings

To ensure that local communities feel supported and informed, it is important to engage in two-way interaction to help foster community connection. The Public Information Officer (PIO) can provide advice on the best ways to engage the impacted community.

A key way of engaging the community is through community meetings. It is important to ensure that any community meetings planned:

- Are led by people with local knowledge, preferably people who are trusted by the community.
- Are structured for two way conversations.
- Use the tools available to properly record issues raised by community members, so you can respond.
- Outcomes from the meeting should be reported to the Incident, Regional and State tiers to add to situational awareness, with a focus on sharing outcomes with councils.

5.20.8 Warning channels

There are numerous methods for providing information and warnings to the community, which include:

- emergency broadcasters (designated community radio stations, ABC radio, local radio, Sky News). For Emergency Warning and Evacuation Warnings, the SEWS can be played prior to the warning being read
- media releases, press conferences and interviews
- social media (Facebook, Twitter)
- VicEmergency website
- VicEmergency app
- VicTraffic
- variable message signs and VicRoads overhead gantry signs
- email distribution lists (local communities and EMTs)
- community meetings and door knocks.

5.20.9 Community Alert Sirens

There are Community Alert Sirens in 40 communities across the State; many in bushfire prone areas.

A siren is triggered through the warnings platform with the issue of any Warning (Watch & Act), Emergency Warning or Prepare to Evacuate or Evacuate Now, unless otherwise requested by the IC. The siren will sound for five minutes to indicate that a significant emergency has been identified and the community should ‘seek further information’.

JSOP04.01 Public Information and Warnings for Class 1 Emergencies provides guidance on community information and warnings before, during and after emergencies.

5.21 Relief (as a part of response)

During and in the immediate aftermath of a bushfire, the relief needs of individuals, families and communities will be complex and specific to each incident. However, there are a number of overarching Relief Priorities for bushfire emergencies which are:

- Provision of need-based assistance for the immediate health and wellbeing of individuals and communities.
- Planned and timely access to restore critical infrastructure (including transport infrastructure, power, and telecommunications).
- Planned and timely return of communities to bushfire impacted areas to minimise further physical and psychological harm.
- Provision of timely, relevant and tailored relief information to assist community members to make informed decisions.
- Effective and efficient state, region/incident and local relief coordination arrangements.
- Coordination of transition from response (including relief) to recovery in partnership with the lead recovery agency and relevant councils.

Some of these priorities support the broader needs of communities beyond the relief phase.

In line with the SEMP, coordination of relief arrangements will be managed across the municipal, regional, and state levels by:

- **Municipal** –councils are responsible for coordinating relief at a local level, which includes establishing relief centres.
- **Regional** – the Department of Families, Fairness and Housing (DFFH) will coordinate relief at the Regional Level.
- **State** – Emergency Recovery Victoria (ERV) will coordinate relief arrangements at the State Level.

At the state level a number of agencies, government departments and non-government organisations have responsibilities for coordinating or providing direct assistance to individuals, families, and communities or indirect assistance through the resupply of essential goods or services to communities isolated in an emergency. Tables 11 and 12 in the SEMP serve as a guide for agencies involved in providing relief, which forms part of the response phase. It details the range and types of assistance, and the providers of each, to support community relief during and immediately after emergencies. State leads are identified the SEMP's Relief Services and Co-ordination table.

5.21.1 Initial Impact Assessment

The Incident Controller (IC) is responsible for initiating an Initial Impact Assessment (IIA). The aim of IIA is to capture, during the initial 48 hours of an emergency, the nature and scale of the bushfire impact on people, community infrastructure, and the economic, natural and built environments, in order for emergency relief and early recovery activities to commence. EMV have published updated Impact Assessment Guidelines in 2022 to assist in the planning and conduct of Impact Assessment (IA) following an emergency event.

The information also informs emergency management agencies and all levels of government of the actual and potential consequences of the bushfire.

To ensure the expedient collection of information, the Incident Controller (IC) may task personnel from any response agency to collect relevant information.

The EMC is responsible for ensuring the coordination, collection, collation and reporting of incident data and impact assessment processes as required. All agencies have a responsibility to assist the EMC with the Impact Assessment process as per the SEMP.

The data from IIAs is used to identify where to focus early recovery activities (including secondary impact assessments) and must be shared immediately with the relief tier coordinators.

5.22 Animal (livestock and companion) and wildlife welfare

If animals are impacted during fire, Victoria's emergency management arrangements provide that DEECA is:

- the primary agency for animal (livestock and companion) welfare (support services); and
- the control agency for wildlife welfare.

Using the Victorian Animal Emergency Welfare Plan, DEECA will establish links with other agencies and organisations with emergency responsibilities as well as those organisations involved in the management of animal welfare to coordinate the delivery of animal welfare support services. They will operate as part of the established emergency response structures and arrangements, quite often commencing in the response phase of a fire.

5.23 Consequence Management

As defined in the EM Act 2013, consequence management means the coordination of agencies, including agencies who engage the skills and services of non-government organisations, which are responsible for managing or regulating services or infrastructure which is, or may be, affected by a major emergency.

Consequence management occurs through the consideration of the wider ramifications of a bushfire. This approach moves the focus from the bushfire to broader consequences affecting a community, regardless of hazard source. Some of the consequences of bushfires to communities that may need to be considered include isolated communities, loss of power, loss of telecommunications, chain supply disruptions, air quality, and economic loss to businesses (particularly tourism-based industries).

Consequence management informs and works in conjunction with relief and recovery activities. It also supports community recovery.

5.24 Transition to Recovery

The SEMP specifies the arrangements for the coordinated planning and management of transition from response to recovery in Victoria.

Transition Plans should be developed collaboratively between Incident Controllers (ICs) and Regional Controllers (RCs), and Recovery Coordinators/Managers at the relevant tiers with appropriate and agreed resources both prior to and post transition. The community must receive continuous services during the transition.

An important component of transition is a seamless transition of communications where relief and recovery messaging should be integrated with response information as early as possible to facilitate a smooth transition to recovery, alongside other components required for effective transition to recovery.

Key concepts guiding transition include:

- Integration of relief and recovery within the Incident Management Team (IMT) – supporting knowledge management into recovery (this is an opportunity to more broadly discuss council presence in the IMT that supports this concept
- Seamless transition of information, impact data and consequence planning
- The use of data from initial impact assessments to identify where to focus early recovery activities and identify resources needed to support immediate recovery needs
- Continuity of relief activities into the recovery phase, if required

Transition from response to recovery is not always a clearly defined step. During large campaign fires there may be a legitimate need to instigate recovery in some areas while the response phase is still in operation. This is a phased transition to recovery. The teams at the relevant incident, regional and State tiers should agree on the timing and phasing of the transition, the activities required and who is responsible.

6. Recovery

The Emergency Recovery Victoria (ERV) has responsibility for the coordination of the activities of agencies following an emergency and ensuring that satisfactory emergency management arrangements are in place to facilitate the recovery from emergencies. In line with the SEMP, recovery tier coordination responsibilities are:

- for state recovery coordination: Emergency Recovery Victoria (ERV)
- for regional recovery coordination: Emergency Recovery Victoria (ERV)
- for municipal recovery coordination: councils

Following a bushfire, the recovery needs of individuals, families and communities will be wide-ranging and complex, however the following recovery priorities may apply:

- The safety, security, physical and mental health of individuals, families and the community, including at risk persons
- The management of debris, waste and physical damage to private assets
- The continuity, restoration, and reconstruction of essential services, critical and community infrastructure to enable the functioning of a community
- Assisting people and communities displaced from their primary place of residence
- The reconnection, re-establishment, and integration of local social and economic systems and networks which provide opportunities for growth, renewal, and innovation
- The establishment and maintenance of effective communication structures to ensure ongoing information flow
- Ensuring community systems and networks are understood, informed, and involved in planning and leading recovery
- Restoration and reconstruction activities to support regeneration of the natural environment and its inhabitants.

The appropriate Recovery Managers/Coordinators should work collaboratively to create event specific recovery plans that are informed by data and information (consider information gathered through response) and an understanding of community context, to build a needs and outcomes based recovery plan where required.

Incident-specific recovery assistance may depend on the scale of the bushfire. Communities should be engaged and supported to lead their recovery. Over time recovery activities will transition into regular mainstream services. The focus will then shift from recovery to community development, renewal and future mitigation planning.

6.1 Public land recovery

DEECA's and Parks Victoria's roles for public land recovery involve:

- rehabilitating and restoring the damage caused by fire control operations
- reopening our strategic road network and protecting it from erosion
- restoring forest and park infrastructure
- protecting water quality and supply
- preventing harm to cultural heritage
- developing and implementing response activities to support ecosystem recovery and regeneration
- helping other agencies with the recovery of the local community when a fire has affected the community's safety or economic or social well-being.

Generally native vegetation will recover from fire over time. How long this takes, depends on how particular ecosystems respond to fire, and the frequency of successive fire events in the same area.

To help some ecosystems recover, rehabilitation and regeneration activities may include:

- controlling invading weeds, whether from seed stock in the soil or from surrounding areas
- controlling or preventing invasions of feral animals
- stabilising areas at risk of soil erosion and replanting them with local indigenous plants
- replanting vegetation damaged by fire control activities.

Control of soil erosion is extra important within water catchments or where sedimentation will otherwise affect river or wetland ecosystems.

Together with Parks Victoria, DEECA target and prioritise treatment of hazardous trees along strategic roads, fire breaks, recreation sites, dams and helipads across Victoria's public land.

6.2 Fences and control lines after bushfires

Landholders are expected to manage risks to their assets from the potential impact of bushfire.

All landholders are expected to have appropriate levels of insurance cover for boundary and internal fences, in the same way any business venture or private householder should protect and insure their assets.

In line with the *Fences Act 1968*, the Victorian Government does not have any ownership or financial stake in fences along the boundary of private and public land except in cases where specific fencing agreements have been entered into with landholder.

It is the responsibility of the owner of private land to fence their property and secure stock within their boundary.

Victoria offers support to the reestablishment of boundary fences between private land and national parks, State parks and State forests, where these fences are destroyed or damaged by bushfire as detailed in the Repair of rural fencing and fire control line and stabilisation policy.

The policy is based on shared responsibility and our government encourages landholders to assess and manage their risks with regard to fire taking up adequate insurance where available.

6.3 Essential water replacement after bushfire

The Victorian Government has a policy regarding replacing essential water used during bushfire fighting operations. This scheme helps to relieve hardship in the community by replenishing essential private water supplies that have been used by fire services during bushfire emergencies.

CFA FFMVic and FRV firefighters have the legislative powers to take water from any waterway or water source for firefighting purposes.

The water owner can request replacement of essential water under this scheme. The water replacement scheme recognises that, while farmers understand the urgency of firefighting, their basic water interests have to be protected too.

If taken for firefighting purposes, a reasonable and sufficient volume of essential water will be provided to sustain the:

- health of affected residences and pets
- health and productivity of stock.

7. Document information

7.1 Acronyms and Terms

Acronyms	Description
AIA	Arrangement for Interstate Assistance
AIDR	Australian Institute for Disaster Resilience
AFAC	Australasian Fire and Emergency Service Authorities Council
AFDRS	Australian Fire Danger Rating System
AIIMS	Australasian Interservice Incident Management System
BAL	Bushfire Attack Level
BMO	Bushfire Management Overlay
BoM	Bureau of Meteorology
BPA	Bushfire Prone Area
BNHCRC	Bushfire and Natural Hazards Cooperative Research Centre (refer to NHRA)
BREAs	Bushfire Risk Engagement Areas
CCOSC	Commissioners and Chief Officers Strategic Committee
CFA	Country Fire Authority
CFA Act 1958	Country Fire Authority Act 1958
Chief Fire Officer	The Chief Fire Officer of the Department of Energy, Environment and Climate Action, who may perform the functions of the Secretary to the Department of Energy, Environment, and Climate Action in respect of fire related activities.
Chief Officer	The Officers-in-Charge of: • Country Fire Authority • Victoria State Emergency Service. Also, any person nominated by an officer in charge above to exercise their powers under the EM Act 2013 s38
Commissioner (FRV)	The Officer-in-Charge of: • Fire Rescue Victoria Also, any person nominated by an officer in charge above to exercise their powers under the EM Act 2013 s38
DCP	Division Command Point
DEECA	Department of Energy, Environment and Climate Action
DFFH	Department of Families, Fairness and Housing
DRA	Dynamic Risk Assessment
DTP	Department of Transport and Planning - includes VicRoads and Rural Roads Victoria

Acronyms	Description
EM	Emergency management
EM Act 2013	*Emergency Management Act 2013*
EMC	Emergency Management Commissioner
EM-COP	Emergency Management Common Operating Picture
EMJPIC	Emergency Management Joint Public Information Committee
EMTs	Emergency Management Teams. Can be at an Incident (IEMTs); Regional (REMTs) or State (SEMT) level.
EMV	Emergency Management Victoria
ERV	Emergency Recovery Victoria
ESTA	Emergency Services Telecommunications Authority
FBAN	Fire Behaviour Analyst
FBI	Fire Behaviour Index
FCV	Field Command Vehicle
FDRs	Fire Danger Ratings
FFMVic	Forest Fire Management Victoria (Agency partnership between DEECA, Parks Victoria, VicForests, Melbourne Water)
FIU	Field Information Update
FMZ	Fire Management Zones
FRV	Fire Rescue Victoria
FRV Act 1958	Fire Rescue Victoria Act 1958 (formerly the Metropolitan Fire Brigades Act 1958)
IAOPs	Interagency Aviation Operating Procedures
IC	Incident Controller
ICC	Incident Control Centre
IIA	Initial Impact Assessment
ICT	Information and Communications Technology
IMT	Incident Management Team
JFMP	Joint Fire Management Program
JSOPs	Joint Standard Operating Procedures - Standard Operating Procedures that are endorsed by control agencies (CFA, FRV, DEECA, VICSES) and approved by EMC - JSOPs are available to agency personnel via EM-COP - See also SOPs
LCF	Local Command Facility
LMAP	Local Mutual Aid Plan

Acronyms	Description
MEMP	Municipal Emergency Management Plan
MEMPC	Municipal Emergency Management Planning Committee
MERC	Municipal Emergency Response Coordinator
MFPO	Municipal Fire Prevention Officer
MOU	Memorandum of Understanding
MRM	Municipal Recovery Manager
NAFC	National Aerial Firefighting Centre
NHRA	Natural Hazards Research Australia
NSP-BPLR	Neighbourhood Safer Places-Bushfire Places of Last Resort
OBRM	Office of Bushfire Risk Management
OH&S	Occupational Safety and Health
PIO	Public Information Officer
RC	Regional Controller
RCC	Regional Control Centre
RCT	Regional Control Team
REMP	Regional Emergency Management Plan
REMPC	Regional Emergency Management Planning Committee
RERC	Regional Emergency Response Coordinator
RSA	Response Support Agency
SAC	State Agency Commander
SCC	State Control Centre
SCRC	State Crisis and Resilience Council
SCT	State Control Team
SEMP	State Emergency Management Plan
SEMT	State Emergency Management Team
SOPs	Standard Operating Procedures - single agencies standard operating procedures - see also JSOPs
SPA	Safe Person Approach
SPLO	Senior Police Liaison Officer
SRC	State Response Controller
SReC	State Recovery Coordinator
SRRS	State Resource Request System
SEWS	Standard Emergency Warning Signal

Acronyms	Description
TFB	Total Fire Ban
TMP	Traffic Management Point
VFRR-B	Victorian Fire Risk Register – Bushfire
VPF	Victorian Preparedness Framework
VTOCS	Victorian Traditional Owner Cultural Fire Strategy

7.2 Linked resources

All hyperlinks were accurate at time of publication. Currency of the linked content remains the responsibility of the host agency.

Linked Resource	Linked Resource
AIDR's Evacuation Planning Handbook	Joint Fuel Management Program
Arrangement for Interstate Assistance	Land use planning and building controls
Australian Disaster Resilience Handbook Collection	Municipal Council role statement
Australian Fire Danger Rating System	Municipal Emergency Management Plans
Australian Warning System	National Aerial Firefighting Centre
Bushfire Management Overlay	National Strategy for Disaster Resilience
Bushfire Management Strategies	Natural Hazards Research Australia
Bushfire Prone Areas	Neighbourhood Safer Places - 'Bushfire Place of Last Resort'
Bushfire Risk Management Report	Office of Bushfire Risk Management
Bushfire Safety Policy Framework	On the Land - Agriculture Fire Management Guidelines
CFA	Participating agencies for mitigation activities specific to bushfire
CFA role statement.	Permits
Code of Practice for Bushfire Management on Public Land	Regional Bushfire Management Strategies
Community Alert Sirens	Regional Emergency Management Plans
Community Fire Refuge	Relief Services and Co-ordination table.
Community Information Guides	Repair of rural fencing and fire control line and stabilisation policy
Control agencies for fire response	Response support agencies, by functional area
DEECA	Seasonal Prohibited Period
DEECA role statement.	SEMP
Emergency Risks in Victoria	SEMP roles and responsibilities
EM-LEARN Lessons Management Framework,	Standard for Managing Exposure to Significant Carbon Monoxide Emissions
EM-Share	State Emergency Management Priorities
ERV	Strategic bushfire management planning
ERV's role statement	Total Fire Ban
FFMVic	Tourism Business Fire Ready Kit
FRV	Victorian Animal Emergency Welfare Plan
FRV role statement	Victorian Building Authority website: private bushfire shelters
Guidelines for Preparing State, Regional and	Victorian Preparedness Framework

Linked Resource	Linked Resource
Municipal Emergency Management Plans	
Information to assist householders and landowners prepare property for bushfire	Victorian Traditional Owner Cultural Fire Strategy